

EZ-USB™ FX2G3

USB 2.0 High-Speed peripheral controller

Overview

EZ-USB™ FX2G3 is a family of USB 2.0 device controllers targeting the established USB 2.0 applications in biometrics, scanner, camera, video and imaging markets. It is based on the MXS40-LP platform with Cortex®-M4 and M0+ MCUs, 512 KB Flash, 128 KB SRAM, 128 KB ROM, Serial Communication Blocks (SCB) and a crypto engine to support various security features. A high bandwidth data subsystem provides DMA data transfers from LVCMOS input to USB output at speeds up to 480 Mbps, USB High-Speed based host systems. A 1024 KB SRAM is included in the high bandwidth data subsystem to provide buffering for data.

FX2G3 will replace the FX2LP USB HS controllers by incorporating most of its features and adding new ones.

Features

- USB interface
 - USB 2.0 HS at 480 Mbps
 - Up to 32 endpoints, 16 IN and 16 OUT; each end point configured as Bulk, Isochronous or Interrupt type
- Dual-core CPU subsystem^[1]
 - 150 MHz Arm® Cortex®-M4F (CM4) CPU with single-cycle multiply, floating point (FP), and memory protection unit (MPU)
 - 100 MHz Arm® Cortex®-M0+ (CM0+) CPU with single-cycle multiply and memory protection unit (MPU)
- Memory subsystem^[1]
 - 512 KB built-in application flash, read-while-write (RWW) support
 - 128 KB SRAM with power and data retention control
 - 128 KB ROM for device initialization, flash write, security, eFuse programming
 - 1 MB SRAM for LVCMOS to USB data buffer
 - 1024 bits one-time programmable (OTP) eFuse array
- General Programmable Interface (GPIF III)
 - LVCMOS parallel-data bus transceiver mode
 - Consists of 16 data, one clock, and 10 control signals
 - 100 MHz SDR in TX and RX mode
- Peripheral IO subsystem – total of 48 shared IOs^[1]
 - Quad SPI (QSPI) configurable as single, dual, quad, dual-quad, and octal interfaces
 - Six Serial Communication Blocks (SCBs) configurable as I2C, UART, or SPI.
 - Pulse-density modulation (PDM) to pulse-code modulation (PCM) converter for microphone
 - One USB Full-Speed (FS) device for virtual communication (COM) function
 - GPIOs: Each peripheral IO can be configured as GPIO

Note

1. Some of these features are available in specific part numbers. See the [Ordering information](#) section for more details.

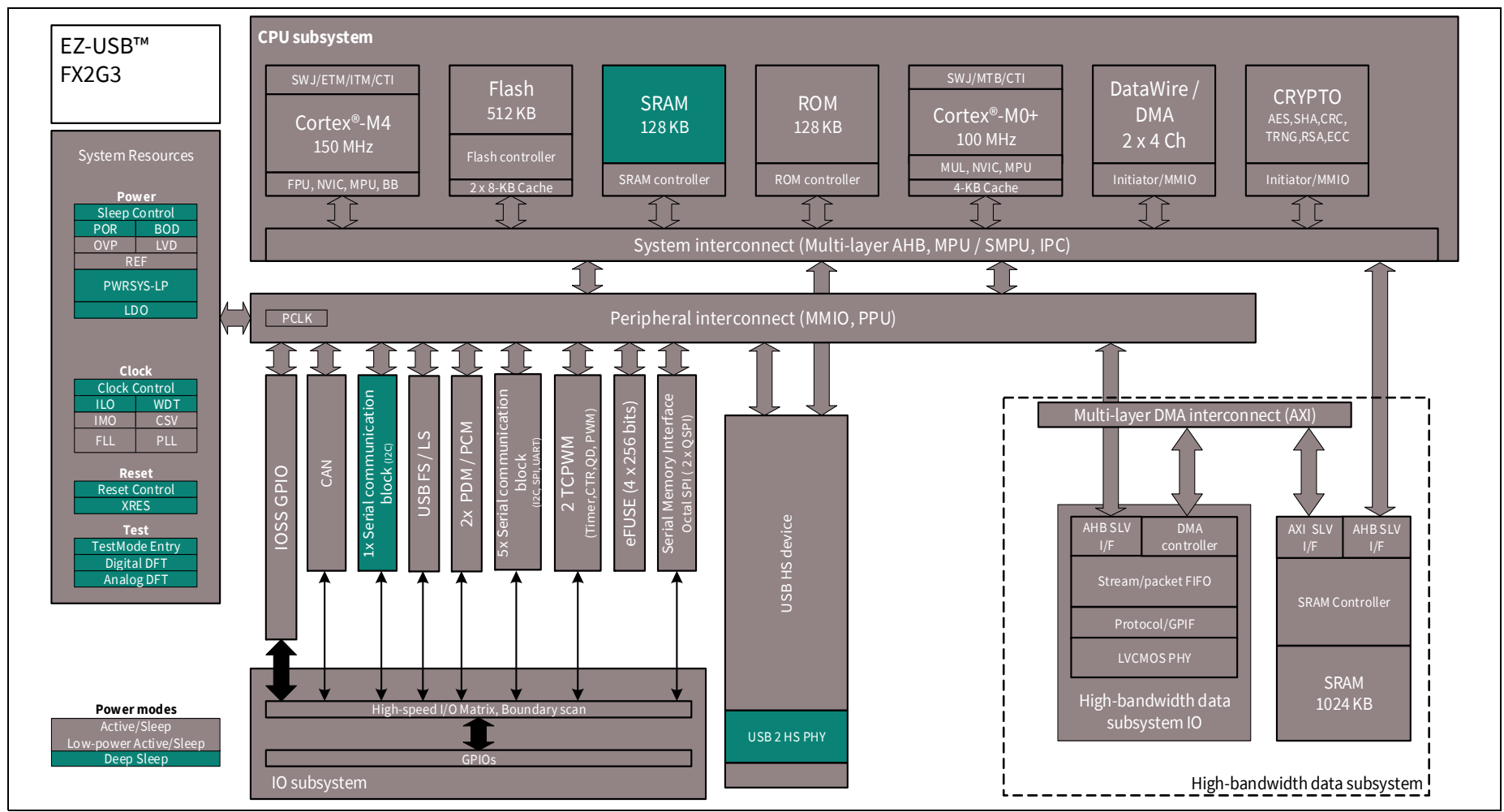
Applications

- Ultra low-power (ULP) with fine-grained power management
 - 1.7 V to 3.6 V operation
 - Deep Sleep mode with SRAM retention
- Flexible clocking options
 - 8MHz internal main oscillator (IMO) with $\pm 2\%$ accuracy
 - Low-power (LP) 32 kHz internal low-speed oscillator (ILO)
 - 24 MHz external crystal oscillator
 - 24 MHz external oscillator
 - Phase-locked loop (PLL) for multiplying clock frequency
 - Frequency-locked loop (FLL) for multiplying IMO frequency
 - Integer and fractional peripheral clock dividers
- Security
 - ROM-based root of trust via uninterrupted "Secure Boot"
 - Stepwise authentication of execution images
 - Secured execution of code in the execute-only mode for protected routines
 - All debug and test ingress paths can be disabled
 - Eight protection contexts
 - Cryptography accelerator
 - Hardware acceleration for symmetric and asymmetric cryptographic methods and hash functions
 - True Random Number Generation (TRNG) function
- Package information
 - 8 mm $\times$ 8 mm 104-pin LGA
- Development tools
 - ModusToolbox™ software enables cross-platform code development with a robust suite of tools and software libraries.
 - Firmware samples including UVC, UAC, HID, CDC, and vendor-specific classes
 - **Infineon Developer Community** enables connection with fellow USB developers around the world, 24 hours a day, and 7 days a week.

Applications

- Secure biometric devices
- Scanners
- Low-resolution cameras
- Industrial automation
- Medical devices
- USB logic analyzer
- USB oscilloscope
- USB data acquisition systems
- USB JTAG debuggers
- USB add-on accessories for mobile phones
- USB protocol analyzer

Block diagram



**EZ-USB™
FX2G3**

System Resources

Power

Sleep Control

PORBOD

OVLVD

REF

PWRSYS-LP

LDO

Clock

Clock Control

ILOWDT

IMOCNV

FLLPLL

Reset

Reset Control

XRES

Test

TestMode Entry

Digital DFT

Analog DFT

Power modes

Active/Sleep

Low-power Active/Sleep

Deep Sleep

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1 EZ-USB™ FX2G3 resources

Table 1 EZ-USB™ FX2G3 resources

Resources	Description
Overview	EZ-USB™ FX2G3 portfolio, EZ-USB™ FX2G3 roadmap
Product selectors	EZ-USB™ FX2G3
Application notes (AN)	Application notes cover a broad range of topics, from basic to advanced level, and include the following: • EZ-USB™ FX2G3 hardware design guidelines • Getting started with EZ-USB™ FX2G3 • Developing UVC + UAC based video capture solution using EZ-USB™ FX2G3
Code examples	Code examples demonstrate product features and usage, and are also available on GitHub repositories.
Reference manuals	Reference manuals provide detailed descriptions of the EZ-USB™ FX2G3 architecture and registers: • EZ-USB™ FX2G3 device controller architecture reference manual • EZ-USB™ FX2G3 device controller register reference manual
EZ-USB™ FX2G3 programming specification	EZ-USB™ FX2G3 programming specification provides the information necessary to program the EZ-USB™ FX2G3 memory.
Development tools	ModusToolbox™ software enables cross-platform code development with a robust suite of tools and software libraries.
Development kits and solution demos	• FX2G3 104LGA Development Kit • EZ-USB™ FX2G3 CAD libraries provide footprint and schematic symbol support for common tools • BSDL files and IBIS models are also available
Infineon Developer Community	Infineon Developer Community enables connection with fellow EZ-USB™ developers around the world, 24 hours a day, 7 days a week, and hosts a dedicated Infineon Community.

2 Functional description

The following sections provide an overview of the features, capabilities, and operation of each functional block identified in the [Block diagram](#). For more information, see the following documentation:

- **Board support package (BSP) documentation**

BSPs are available on [GitHub](#). They are aligned with Infineon kits and provide files for basic device functionality such as hardware configuration files, startup code, and linker files. The BSP also includes other libraries that are required to support a kit. Each BSP has its own documentation, but typically includes an API reference such as the example here. This search link finds all currently available BSPs on the Infineon [GitHub](#) site.

- **Peripheral driver library (PDL) Application Programming Interface (API) reference manual**

The PDL integrates device header files and peripheral drivers into a single package and supports all EZ-USB™ FX2G3 product lines. The drivers abstract the hardware functions into a set of easy-to-use APIs. These are fully documented in the [PDL API Reference](#). Example applications that use the EZ-USB™ FX2G3 PDL download it automatically from the GitHub repository.

- **Architecture reference manual**

The architecture reference manual provides a detailed description of each resource in the device. This is the next reference to use if it is necessary to understand the operation of the hardware below the software provided by PDL. It describes the architecture and functionality of each resource and explains the operation of each resource in all modes. It provides specific guidance regarding the use of associated registers.

- **Register reference manual**

The register reference manual provides a complete list of all registers in the device. It includes the breakdown of all register fields, their possible settings, read/write accessibility, and default states. All registers that have a reasonable use in typical applications have functions to access them from within PDL. Note that ModusToolbox™ and PDL may provide software default conditions for some registers that are different from and override the hardware defaults.

2.1 USB interface

2.1.1 USB 2.0 High-Speed device

The USB HS device will be compliant to USB 2.0 specifications.

The following features are supported:

- EZ-USB™ FX2G3 is pre-programmed with a bootloader which allows firmware download over USB
- The bootloader will support boot mode selection using PMODE GPIO
- Supports firmware download to an external flash using I2C/SPI/QSPI interface
- Supports USB bus-powered and self-powered mode
- Detect USB-C connector orientation using CC lines. External Rd resistors are required on CC lines.

2.1.2 Compliance to USB 2.0 DC resistance ECN

The ECN updates the USB 2.0 specification to account for DC resistance (DCR) on the D+ and D– data paths between the USB transceiver device and the connector. This increased resistance could be due to multiple crossbar switches and/or passive components such as common-mode chokes. The ECN updates affect the following High-Speed parameters: VHSOH transmit voltage, VHSDSC disconnect level, and VHSSQ squelch level.

2.1.2.1 Allowable transmit output voltage (VHSOH)

The ECN expands the allowable TX output voltage from 400 mV ± 10% to 360 mV to 625 mV. EZ-USB™ FX2G3 supports the default transmit voltage of 400 mV and a programmable nominal range of 370 mV to 520 mV.

2.1.2.2 Disconnect envelope detector threshold voltage (VHSDSC)

The ECN changes the disconnect envelope detector threshold range from 525 mV to 625 mV to a much wider range. The design guide shows a minimum threshold range of 550 mV to 640 mV to a maximum of ~820 mV to ~950 mV. EZ-USB™ FX2G3 supports programmable threshold levels that cover the original 525 mV to 625 mV range up to the maximum range of ~820 mV to ~950 mV.

2.1.2.3 Squelch threshold voltage (VHSSQ)

The ECN changes the squelch threshold from 100 mV to 150 mV to a lowered range depending on the DCR. FX2G3 supports the original threshold range of 100 mV to 150 mV and can be programmed to a lowered threshold range of 85 mV to 135 mV.

FX2G3 also supports multiple programmable options for pre-emphasis which can be used to compensate for increased DCR and other components that cause channel degradation.

2.1.3 Renumeration

EZ-USB™ FX2G3 has a soft configuration, where one chip can take on the identities of multiple distinct USB devices. When first plugged into USB, EZ-USB™ FX2G3 enumerates automatically with the vendor ID (0x04B4) and downloads firmware and USB descriptors over the USB interface. The downloaded firmware executes an electrical disconnect and reconnect. EZ-USB™ FX2G3 enumerates again, this time as a device defined by the downloaded information. This patented two-step process, called “ReNumeration”, happens instantly when the device is plugged in.

2.2 High-bandwidth data subsystem

The high bandwidth data subsystem provides DMA data transfers from LVCMOS subsystem to USB 2 Device at speeds up to 480 Mbps based host systems. A 1024 KB SRAM is included in the high bandwidth data subsystem to provide buffering for data. The data is transferred over an AXI 3 interconnect supporting 480 Mbps operation. The subsystem interfaces to the System Interconnect for data transfer to other low bandwidth peripherals and the MMIO Interconnect for configuration.

The high-bandwidth data subsystem IO is a LVCMOS parallel interface.

2.2.1 LVCMOS parallel interface

The LVCMOS parallel interface supports RX and TX interface. It has 16 LVCMOS IOs and 10 control signals. The maximum data rate on each LVCMOS IO is:

- 100 MHz SDR only in TX and RX mode

The maximum frequency that the control signals can operate is up to 50 MHz

2.2.2 GPIF III interface

The GPIF III is a programmable state machine that enables a flexible interface that may function as a master or slave in industry-standard or proprietary interfaces. Only parallel interfaces are implemented with GPIF III.

The following are the GPIF III features:

- Functions as master or slave
- Provides 256 firmware programmable states
- Supports 8-bit and 16-bit parallel data bus

The GPIF III state transitions are based on control input signals. The control output signals are driven as a result of the GPIF III state transitions. For more details, see the architecture reference manual.

2.2.3 High-bandwidth direct memory access (HB-DMA)

The DMA controller includes the following features:

- Supports multiple simultaneous streams
- Supports maximum throughput of 480 Mbps
- DMA is configurable by the CPU. Once configured, the DMA operates automatically and transfers the video stream from the LVCMOS interface to the USB device without CPU intervention.
- Adds headers and footers with every data packet (payload) without CPU intervention
- Ability to wrap the USB packet at the end of frame using control signal and also update the header / footer without CPU intervention

2.3 CPU and memory subsystem

The CPU subsystem has the following major components:

- 150 MHz Cortex®-M4 core
- 100 MHz Cortex®-M0+ core
- 512 KB flash memory
- 128 KB SRAM
- 128 KB ROM
- DMA
- Cryptography accelerator

Following are the multiple bus masters in EZ-USB™ FX2G3 (see **“Block diagram”** on page 3):

- CPUs
- DMA controllers
- QSPI
- USB
- Cryptography accelerator

Generally, all the memory and peripherals can be accessed and shared by all bus masters through multi-layer Arm® AMBA high-performance bus (AHB) arbitration. Accesses between CPUs can be synchronized using an inter-processor communication (IPC) block.

2.3.1 CPUs

There are two Arm® Cortex® CPUs:

1. The Cortex®-M4 (CM4) has a single-cycle multiply, a floating point unit (FPU), and a memory protection unit (MPU). It can run up to 150 MHz. This is the main CPU, designed for a short interrupt response time, high code density, and high throughput.
CM4 implements a version of the Thumb instruction set based on Thumb-2 technology (defined in the [Arm®v7-M Architecture Reference Manual](#)).
2. The Cortex®-M0+ (CM0+) has a single-cycle multiply and an MPU. It can run at up to 100 MHz; however, for CM4 speeds above 100 MHz, CM0+ and bus peripherals are limited to half the speed of CM4. Therefore, for CM4 running at 150 MHz, CM0+ and peripherals are limited to 75 MHz.

CM0+ is the secondary CPU; it is used to implement system calls and device-level safety and protection features. CM0+ provides a secured, uninterruptible boot function. This helps ensure that post boot, system integrity is checked and memory and peripheral access privileges are enforced.

CM0+ implements the Arm®v6-M Thumb instruction set (defined in the [Arm®v6-M Architecture Reference Manual](#)).

The CPUs have the following power draw, at VDDD = 3.3 V:

Table 2 Active current slope at VDDD = 3.3 V, using the internal LDO regulator

System power consumption	
CPU	LP mode
Cortex®-M0+	28 µA/MHz
Cortex®-M4	40 µA/MHz

The CPUs can be selectively placed in their Sleep and Deep Sleep power modes as defined by Arm®.

Both CPUs have nested vectored interrupt controllers (NVIC) for rapid and deterministic interrupt response and wakeup interrupt controllers (WIC) for CPU wakeup from Deep Sleep power mode.

The CPUs have extensive debug support. EZ-USB™ FX2G3 has a debug access port (DAP) that acts as the interface for device programming and debugging. An external programmer or debugger (the “host”) communicates with the DAP through the device serial wire debug (SWD) or Joint Test Action Group (JTAG) interface pins. Through the DAP (and subject to restrictions), the host can access the device memory and peripherals as well as the registers in both CPUs.

Each CPU offers the following debug and trace features:

- CM4 supports
 - Six hardware breakpoints and four watchpoints
 - 4-bit embedded trace macrocell (ETM)
 - Serial wire viewer (SWV)
 - Printf()-style debugging through the single wire output (SWO) pin
- CM0+ supports
 - Four hardware breakpoints and two watchpoints
 - A micro trace buffer (MTB) with 4 KB dedicated RAM

Additionally, EZ-USB™ FX2G3 has an embedded cross trigger for synchronized debugging and tracing of both CPUs.

2.3.2 Interrupts

EZ-USB™ FX2G3 has 168 system and peripheral interrupt sources, and supports interrupts and system exceptions on both CPUs. CM4 has 168 interrupt request lines (IRQ), with the interrupt source 'n' directly connected to IRQn. CM0+ has eight interrupts IRQ[7:0] with configurable mapping of one or more interrupt sources to any of the IRQ[7:0]. CM0+ also supports eight internal (software-only) interrupts.

Each interrupt supports configurable priority levels (eight levels for CM4 and four levels for CM0+). Up to four system interrupts can be mapped to each of the CPUs' non-maskable interrupts (NMI). Up to 39 interrupt sources are capable of waking the device from Deep Sleep power mode using the WIC. See the EZ-USB™ FX2G3 architecture manual for more details.

2.3.3 Inter-processor communication (IPC)

In addition to the Arm® SEV and WFE instructions, a hardware IPC block is included. It includes 16 IPC channels and 16 IPC interrupt structures. The IPC channels can be used to implement data communication between the processors. Each IPC channel also implements a locking scheme which can be used to manage shared resources. The IPC interrupts let one processor interrupt the other, signaling an event. This is used to trigger events such as notify and the release of the corresponding IPC channels. Some IPC channels and other resources are reserved, as [Table 3](#) shows.

Table 3 Distribution of IPC channels and other resources

Resource available	Resources consumed
IPC channels, 16 available	Eight reserved
IPC channels, 16 available	Eight reserved
Other interrupts	One reserved
CM0+ NMI	Reserved
Other resources: clock dividers, DMA channels, and so on	One CM0+ interrupt MUX

2.3.4 Direct memory access (DMA) controllers

EZ-USB™ FX2G3 has three DMA controllers, which support CPU-independent access to memory and peripherals. Two of them have 29 channels each and the third has four channels. The descriptors for DMA channels can be in SRAM or flash. Therefore, the number of descriptors is limited only by the size of the memory. Each descriptor can transfer data in two nested loops with configurable address increments to the source and destination. The size of data transfer per descriptor varies based on the type of DMA channel. See the EZ-USB™ FX2G3 architecture manual for more details.

2.3.5 Cryptography accelerator

This subsystem consists of hardware implementation and acceleration of cryptographic functions and random number generators.

The cryptography subsystem supports the following:

- Encryption/decryption functions
 - Data encryption standard (DES)
 - Triple DES (3DES)
 - Advanced encryption standard (AES) (128-, 192-, 256-bit)
 - Elliptic curve cryptography (ECC)
 - RSA cryptography functions
- Hashing functions
 - Secure Hash Algorithm (SHA)
 - SHA-1
 - SHA-224/-256/-384/-512
- Message authentication functions (MAC)
 - Hashed message authentication code (HMAC)
 - Cipher-based message authentication code (CMAC)
- 32-bit cyclic redundancy code (CRC) generator
- Random number generators
 - Pseudo random number generator (PRNG)
 - True random number generator (TRNG)

2.3.6 Protection units

This product line has multiple types of protection units to control erroneous or unauthorized access to memory and Peripheral registers. CM4 and CM0+ have Arm® MPUs for protection at the bus master level. Other bus masters use additional MPUs. Shared memory protection units (SMPUs) help implement protection for memory resources that are shared among multiple bus masters. The peripheral protection units (PPU) are similar to SMPUs but are designed for protecting the peripheral register space. Protection units support memory and peripheral access attributes including address range, read/write, code/data, privilege level, secured/non-secured, and protection context. The protection units are configured at boot to control access privileges and rights for bus masters and peripherals. Up to eight protection contexts (boot is in protection context 0) allow access privileges for memory and system resources to be set by the boot process per protection context by bus master and code privilege level. Multiple protection contexts are available.

2.3.7 Memory

EZ-USB™ FX2G3 contains flash, SRAM, ROM, and eFuse memory blocks.

2.3.7.1 Flash

There is up to 512 KB of internal application flash. There are also two 32 KB flash sectors:

- Auxiliary flash (AUXflash): used for EEPROM emulation
- Supervisory flash (Sflash): Data stored in Sflash includes device trim values, **Flash boot** code, and encryption keys. After the device transitions into the “Secure” lifecycle stage, Sflash can no longer be changed.

The flash has 128-bit-wide accesses to reduce power. Write operations can be performed at the row level. A row is 512 bytes. The flash controller has two caches, one for each CPU. Each cache is 8 KB, with 4-way set associativity.

2.3.7.2 SRAM

Up to 128 KB of SRAM is provided. The SRAM bank provides control over power modes to manage power consumption. Power control and retention granularity are configurable in four 32 KB regions. For normal operation, the SRAM regions can be enabled or disabled to save power. For Deep Sleep mode, the regions can also be configured to retain data.

2.3.7.3 ROM

The 128 KB ROM, also referred to as the supervisory ROM (SROM), provides code (**ROM boot**) for several system functions. The ROM contains device initialization, flash write, security, eFuse programming, and other system-level routines. ROM code is executed only by the CM0+ CPU, in protection context 0. A system function can be initiated by either CPU or the DAP. This causes an NMI in CM0+, which causes CM0+ to execute the system function.

2.3.7.4 eFuse

A one-time-programmable (OTP) eFuse array consists of 1024 bits, of which 648 bits are reserved for system use such as die ID, device ID, initial trim settings, device life cycle, and security settings. The remaining bits are available for storing key information, hash values, unique IDs, or similar custom content.

Each fuse is individually programmed; once programmed (or “blown”), its state cannot be changed. Blowing a fuse transitions it from the default state of ‘0’ to ‘1’. To program an eFuse, VDDD must be at $2.5\text{ V} \pm 5\%$, at 14 mA.

Because blowing an eFuse is an irreversible process, programming is recommended only in mass production under controlled factory conditions. For more information, see the EZ-USB™ FX2G3 programming specifications.

2.3.8 Boot code

Two blocks of code, ROM boot, and flash boot are preprogrammed into the device and work together to provide device startup and configuration, basic security features, lifecycle stage management, and other system functions.

2.3.8.1 ROM boot

On a device reset, the boot code in ROM is the first code to execute. This code performs the following:

- Integrity checks of flash boot code
- Device trim setting (calibration)
- Setting the device protection units (DPU)
- Setting device access restrictions for the “Secure” lifecycle states ROM cannot be changed and acts as the root of trust in a secured system

2.3.8.2 Flash boot

Flash boot is firmware stored in SFlash that ensures that only a validated application may run on the device. It also ensures that the firmware image has not been modified, such as by a malicious third party.

Flash boot:

- Is validated by ROM boot
- Runs after ROM boot and before the user application
- Enables system calls
- Configures the DAP
- Launches the user application

If the user application cannot be validated, then flash boot ensures that the device is transitioned into a safe state.

2.3.9 Memory map

Both CPUs have a fixed address map, with shared access to memory and peripherals. The 32-bit (4 GB) address space is divided into the Arm®-defined regions shown in [Table 4](#). Note that code can be executed from the code and external RAM regions.

Table 4 Address map for CM4 and CM0+

Address range	Name	Use
0x0000 0000 to 0x1FFF FFFF	Code	Program code region. Data can also be placed here. It includes the exception vector table, which starts at address 0.
0x2000 0000 to 0x3FFF FFFF	Reserved	Reserved
0x4000 0000 to 0x5FFF FFFF	Peripheral	All peripheral registers. Code cannot be executed from this region. CM4 bit-band in this region is not supported in EZ-USB™ FX2G3.
0x6000 0000 to 0x9FFF FFFF	External RAM	Quad SPI, (see the “ Quad SPI/serial memory interface (SMIF) ” on page 24 section). Code can be executed from this region.
0xA000 0000 to 0xDFFF FFFF	External Device	Not used
0xE000 0000 to 0xE00F FFFF	Private Peripheral Bus	Provides access to peripheral registers within the CPU core
0xE010 0A000 to 0xFFFF FFFF	Device	Device-specific system registers

The device memory map shown in [Table 5](#) applies to both CPUs. That is, the CPUs share access to all EZ-USB™ FX2G3 memory and peripheral registers.

Table 5 Internal memory address map for CM4 and CM0+

Address range	Memory type	Size
0x0000 0000 to 0x0001 FFFF	ROM	128 KB
0x0800 0000 to 0x0801 FFFF	SRAM	128 KB
0x1000 0000 to 0x1007 FFFF	Internal application flash	Up to 512 KB
0x1400 0000 to 0x1400 7FFF	Auxiliary flash, can be used for EEPROM Emulation	32 KB
0x1600 0000 to 0x1600 7FFF	SFlash	32 KB
0x1C00 0000 to 0x1C0F FFFF	High bandwidth DMA buffer SRAM	Up to 1 MB

Note that EZ-USB™ FX2G3 SRAM is located in the Arm® Code region for both CPUs (see [Table 4](#)). There is no physical memory located in the CPUs' Arm® SRAM regions. Only the data that is placed in the DMA Buffer SRAM can be transferred across the USB HS interface. This region can be used for both data and volatile code segments.

2.4 System resources

2.4.1 Power system

The power system assures that voltage levels are as required for each respective mode and will either delay mode entry (for example, on power-on reset (POR)) until voltage levels are as required for proper function or generate resets (brown-out detect (BOD)) when the power supply drops below specified levels. The design guarantees safe chip operation between the power supply voltage drops below-specified levels (for example, below 1.7 V) and the reset occurs. There are no voltage sequencing requirements.

The VDDD supply (1.7 V to 3.6 V) powers a low-dropout regulator (LDO). The system low-power (LP) operates the core operating voltage (VCCD) at 1.1 V and offers high-performance with no restrictions on device configuration.

2.4.2 Power modes

EZ-USB™ FX2G3 can operate in four systems and three CPU power modes. These modes are intended to minimize the average power consumption in an application^[2].

The EZ-USB™ FX2G3 supports the following power modes, in order of decreasing power consumption:

- System low-power (LP) – All peripherals and CPU power mode are available at maximum speed
- CPU Active – CPU is executing code in system LP
- CPU Sleep – CPU code execution is halted in system LP
- CPU Deep Sleep – CPU code execution is halted and system Deep Sleep is requested in system LP mode
- System Deep Sleep – Only low-frequency peripherals are available after both CPUs enter CPU Deep Sleep mode
- Normal mode: This is the full-functional operating mode. The internal CPU clock and the internal PLLs are enabled in this mode.
 - USB 2.0 operation at 480 Mbps
 - High-bandwidth data subsystem IOs port in LVCMOS SDR mode
- Other Low-power modes (see [Table 6](#)):
 - Suspend mode
 - Deep Sleep mode
 - Core power-down mode

CPU Active, Sleep, and Deep Sleep are standard Arm®-defined power modes supported by the Arm® CPU instruction set architecture (ISA). The system LP and Deep Sleep modes are additional low-power modes supported by the EZ-USB™ FX2G3.

Note

2. The power consumption depends on how the EZ-USB™ FX2G3 IOs are utilized in the application.

Functional description

Table 6 Entry and exit methods for low-power modes

Low-power mode	Characteristics	Methods of entry	Methods of exit
Suspend mode	- The power consumption in this mode does not exceed 1.8 mW - All I/Os maintain their previous state - The power supply for the wakeup source and core power must be retained. All other power domains can be turned ON/OFF individually. - CPU shall be in Deep Sleep mode - The states of the configuration registers, buffer memory, and all internal RAM are maintained - All transactions must be completed before EZ-USB™ FX2G3 enters Suspend mode (the state of outstanding transactions is not preserved) - The firmware resumes operation from where it was suspended (except when woken up by an XRES assertion) because the program counter does not reset	Firmware executing on the Arm® core can put EZ-USB™ FX2G3 into Suspend mode. For example, on a USB suspend condition, the firmware may decide to put EZ-USB™ FX2G3 into Suspend mode.	- D+ transitioning to LOW or HIGH - D– transitioning to LOW or HIGH - Detection of VBUS - GPIO or I2C activity - GPIF III interface assertion of any LVCMOS CTL pins - Assertion of XRES pins

Table 6 Entry and exit methods for low-power modes (continued)

Low-power mode	Characteristics	Methods of entry	Methods of exit
Deep Sleep mode	- The power consumption in this mode does not exceed 120 μW - All configuration register settings and program/data RAM contents are preserved - All blocks are powered OFF including the USB PHY - GPIO pins maintain their configuration - Crystal oscillator is turned off - Internal PLL is turned off - USB transceiver is turned off - Arm® core is in the deep sleep state. Upon wakeup, the core re-starts and runs the program stored in the program/data RAM. - Power supply for the wakeup source and core power must be retained. All other power domains can be turned on/off individually.	Firmware executing on Arm® core configures the appropriate register	GPIO or I2C activity - GPIF III interface assertion of LVCMOS I/Os - Assertion of XRES pins
Core power-down mode	- The power consumption in this mode does not exceed 15 μW - Core power is turned off along with all other power supplies to EZ-USB™ FX2G3 - All buffer memory, configuration registers and the program RAM do not maintain their state. After exiting this mode, the firmware will be reloaded - In this mode, all other power domains can be turned on/off individually.	Turn off all power domains	Reapply power supplies and assertion of XRES pins

2.4.3 Booting options

EZ-USB™ FX2G3 can load boot images from various sources, selected by the configuration of the PMODE pin. Following are the EZ-USB™ FX2G3 boot options:

- Boot from USB
- Boot from internal application flash

Table 7 EZ-USB™ FX2G3 booting options

PMODE_P13.0	Boot from
LOW	Internal application flash
HIGH	USB

Functional description

2.4.4 Clock system

The clock system as shown in **Figure 1**, consists of the following features:

- Internal main oscillator (IMO)
- Internal low-speed oscillator (ILO)
- External 24-MHz crystal oscillator (ECO)
- External clock input
- Two phase-locked loops (PLLs)
- One frequency-locked loop (FLL)

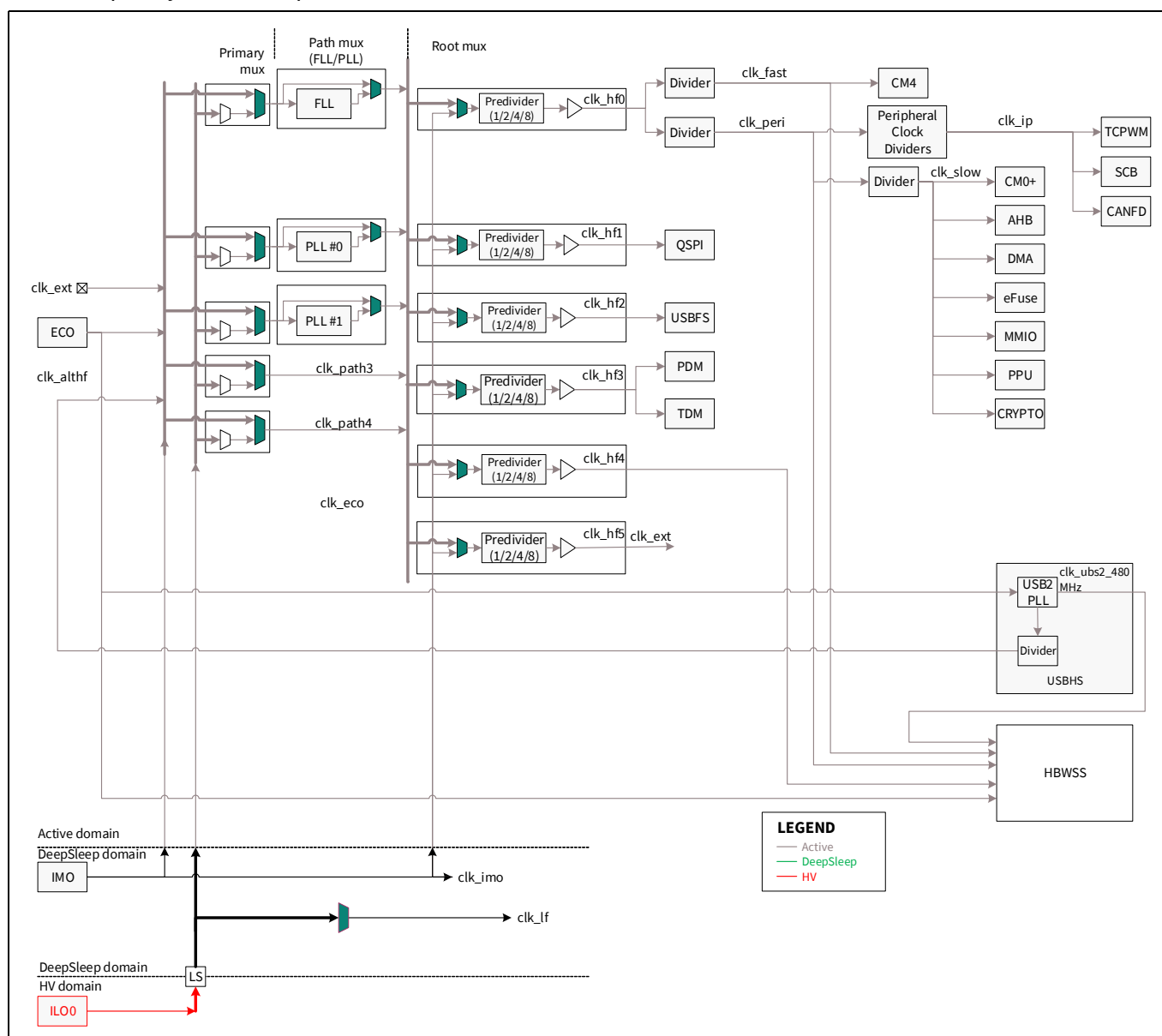


Figure 1 Clocking diagram

2.4.5 Internal main oscillator (IMO)

The IMO is the primary source of internal clocking. It is trimmed at the factory to achieve the specified accuracy. The IMO frequency is 8 MHz and the tolerance is $\pm 2\%$.

2.4.6 Internal low-speed oscillator (ILO)

The ILO is a very low-power oscillator, nominally 32 kHz, which operates in all power modes. The ILO can be calibrated against a higher accuracy clock for better accuracy.

2.5 System-level ESD

All pins of EZ-USB™ FX2G3 support ± 2.2 -kV human body model (HBM) based on JESD22-A114 Specification. IEC protection has to be provided at the system level.

2.5.1 External crystal oscillators (ECO)

When an external crystal is used, the start-up time for the clock output from the crystal driver circuit is determined by the following:

- External crystal oscillator's intrinsic characteristics
- Internal settings of the crystal driver
- Load capacitance on the XIP and XOP pins

Figure 2 shows the crystal model components, the load capacitors, the XIP and XOP pins of EZ-USB™ FX2G3, and the crystal driver circuit. The model consists of the following parameters:

- Crystal motional capacitance (C_s)
- Crystal motional inductance (L_s)
- Crystal equivalent series resistance (ESR)
- Crystal shunt capacitance (C_0)
- Load capacitors CL1 and CL2

Refer the **“Clock system”** on page 20 for the crystal values and the crystal datasheet for the load capacitor values. The ECO requires balanced external load capacitors.

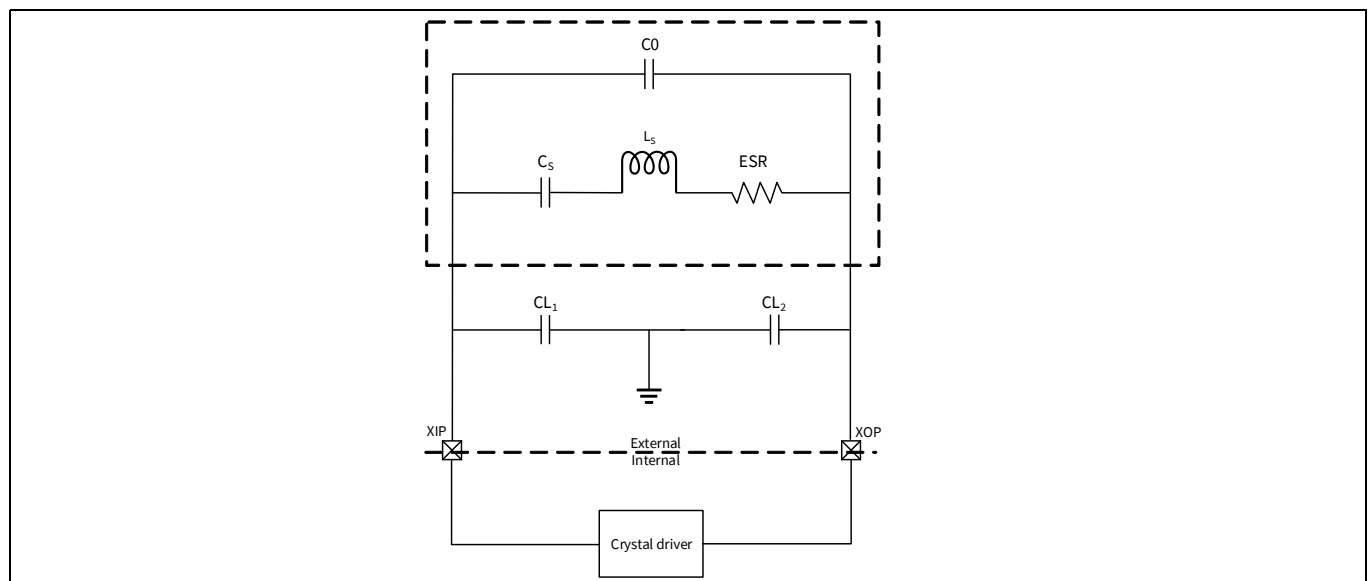


Figure 2 Oscillator circuits

2.5.2 Clock dividers

Integer and fractional clock dividers are provided for peripheral use and timing purposes. There are:

- Eight 8-bit clock dividers
- Sixteen 16-bit integer clock dividers
- Four 16.5-bit fractional clock dividers
- One 24.5-bit fractional clock divider

2.5.3 Trigger routing

EZ-USB™ FX2G3 contains a trigger multiplexer block. This is a collection of digital multiplexers and switches that are used for routing trigger signals between peripheral blocks and between GPIOs and peripheral blocks.

There are two types of trigger routing. Trigger multiplexers have reconfigurability in the source and destination. There are also hardwired switches called “one-to-one triggers”, which connect a specific source to a destination. The user can enable or disable the route.

2.5.4 Reset

EZ-USB™ FX2G3 can be reset from a variety of sources:

- POR to hold the device in reset while the power supply ramps up to the level required for the device to function properly. POR activates automatically at power-up
- BOD reset to monitor the digital voltage supply VDDD and generate a reset if VDDD falls below the minimum required logic operating voltage
- External reset dedicated pin (XRES) to reset the device using an external source. The XRES pin is active LOW and has a weak internal pull up of 85 kΩ. An optional 4.7 kΩ pull up resistor can be connected to VDDD as shown in [Figure 3](#).

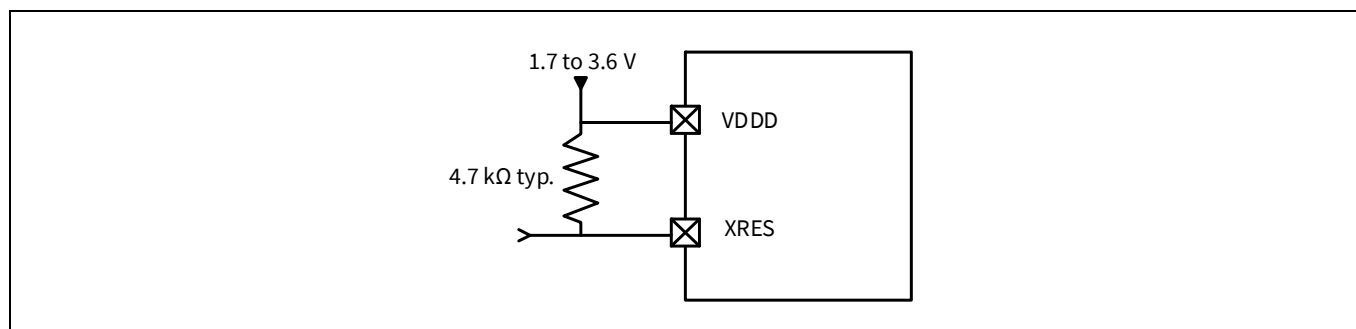


Figure 3 XRES connection diagram

- Software-initiated reset to reset the device on demand using the firmware
- Logic-protection fault can trigger an interrupt or reset the device if unauthorized operating conditions occur; for example, reaching a debug breakpoint while executing privileged code

The reset events are asynchronous and guarantee reversion to a known state. Some of the reset sources are recorded in a register, which is retained through reset and allows the software to determine the cause of the reset. RAM contents are not retained across the reset event.

2.6 Fixed-function digital

2.6.1 Timer/Counter/Pulse-width Modulator (TCPWM)

- The TCPWM supports the following operational modes:
 - Timer-counter with compare
 - Timer-counter with capture
 - Quadrature decoding
 - Pulse-width modulation (PWM)
 - Pseudo-random PWM
 - PWM with dead time
- Up, down, and up/down counting modes
- Clock prescaling (division by 1, 2, 4,... 64, 128)
- Double buffering of compare/capture and period values
- Underflow, overflow, and capture/compare output signals
- Supports interrupt on:
 - Terminal count – Depends on the mode; typically occurs on overflow or underflow
 - Capture/compare – The count is captured to the capture register or the counter value equals the value in the compare register
- Complementary output for PWMs
- Selectable start, reload, stop, count, and capture event signals for each TCPWM; with the rising edge, falling edge, both edges, and level trigger options. The TCPWM has a Kill input to force outputs to a predetermined state.

In this device, there are 2×32 -bit TCPWMs.

2.6.2 Serial Communication Block (SCB)

EZ-USB™ FX2G3 has six SCBs (see [Table 9](#)).

- **I²C mode:** The SCB can implement a full multi-master and slave interface (it is capable of multi-master arbitration). This block can operate at speeds of up to 1 Mbps (Fast Mode Plus). It also supports EZI2C, which creates a mailbox address range and effectively reduces I2C communication to reading from and writing to an array in memory. The SCB supports a 256-byte FIFO for receiving and transmitting. The I2C peripheral is compatible with I2C standard-mode, Fast Mode, and Fast Mode Plus devices as defined in the NXP I2C-bus specification and user manual (UM1024). The I2C bus I/O is implemented with GPIO in open-drain modes. SCB0, SCB1, SCB4, SCB5, and SCB6 support I2C mode.
- **UART mode:** This is a full-featured UART operating at up to 8 Mbps. It supports automotive single-wire interface (LIN), infrared interface (IrDA), and SmartCard (ISO 7816) protocols, all of which are minor variants of the basic UART protocol. In addition, it supports the 9-bit multiprocessor mode that allows the addressing of peripherals connected over common RX and TX lines. Common UART functions such as parity error, break detect, and frame error are supported. A 256-byte FIFO allows much greater CPU service latencies to be tolerated. SCB0, SCB1, SCB4, and SCB6 support UART mode.
- **SPI mode:** The SPI mode supports full Motorola SPI, TI Secure Simple Pairing (SSP) (which essentially adds a start pulse that is used to synchronize SPI codecs), and National Microwire (a half-duplex form of SPI). The SPI block supports an EZSPI mode in which the data interchange is reduced to reading and writing an array in memory. The SPI interface operates with a 25-MHz clock. SCB2 and SCB5 support SPI mode.

2.6.3 USB Full-Speed device interface

This product line incorporates a Full-Speed USB device interface. The device can have up to eight endpoints. A 512-byte SRAM buffer is provided and DMA is supported.

Note that if the USB-FS pins are not used, connect VDDUSB to ground and leave the USB2DP2 and USB2DN2 pins unconnected.

2.6.4 Quad SPI/serial memory interface (SMIF)

A serial memory interface is provided, running at up to 80 MHz. It supports single, dual, quad, dual-quad, and octal SPI configurations and supports up to four external memory devices. It supports two modes of operation:

- MMIO, a command mode interface that provides data access via registers and FIFOs
 - Execute-In-place (XIP), in which AHB reads and writes are directly translated to SPI read and write transfers
- In XIP mode, the external memory is mapped into the EZ-USB™ FX2G3 internal address space, enabling code execution directly from the external memory. To improve performance, a 4 KB cache is included. XIP mode also supports AES-128 on-the-fly encryption and decryption, enabling secured storage and access of code and data in the external memory.

2.7 GPIO

EZ-USB™ FX2G3 has 4 GPIOs including the pins used for low-speed peripheral interface, which implement the following:

- Eight drive modes:
 - Input only
 - Weak pull-up with strong pull-down
 - Strong pull-up with weak pull-down
 - Open drain with strong pull-down
 - Open drain with strong pull-up
 - Strong pull-up with strong pull-down
 - Weak pull-up with weak pull-down
 - In strong drive mode, there are four selectable drive strengths for IOL and IOH current drive
- Input threshold select (CMOS or LVTTL)
- Selectable slew rates for dV/dt-related noise control to improve EMI

The pins are organized into logical entities called ports, which are up to eight pins wide. Data output and pin state register store, respectively, the values to be driven on the pins and the input states of the pins.

Every pin can generate an interrupt if enabled, and each port has an interrupt request (IRQ) associated with it.

The ports 10.0 and 10.1 pins are capable of overvoltage tolerant (OVT) operation, where the input voltage may be higher than VDDD. OVT pins are commonly used with I2C to allow powering the chip OFF while maintaining a physical connection to an operating I2C bus without affecting its functionality.

GPIO pins can be ganged to source or sink higher values of current. GPIO pins, including OVT pins, may not be pulled up higher than the absolute maximum; see **“Electrical specifications”** on page 37.

2.8 Special-function peripherals

2.8.1 Audio subsystem

This subsystem consists of PDM to PCM decoder channel.

The PDM-to-PCM decoder implements a single hardware RX FIFO that decodes a stereo or mono 1-bit PDM input stream to PCM data output. The following features are supported:

- Programmable data output word length – 16/18/20/24 bits
- Programmable gain amplifier (PGA) for volume control from –12 dB to +10.5 dB in 1.5 dB steps
- Configurable PDM clock generation. Range from 384 kHz to 3.072 MHz.
- Droop correction and configurable decimation rate for sampling; up to 48 ksps
- Programmable high-pass filter gain
- Interrupt mask events – not empty, overflow, trigger, and underflow
- Configurable FIFO trigger level with DMA support

The PDM-to-PCM decoder is commonly used to connect to digital PDM microphones. Up to two microphones can be connected to the same PDM Data line.



3 Pinouts

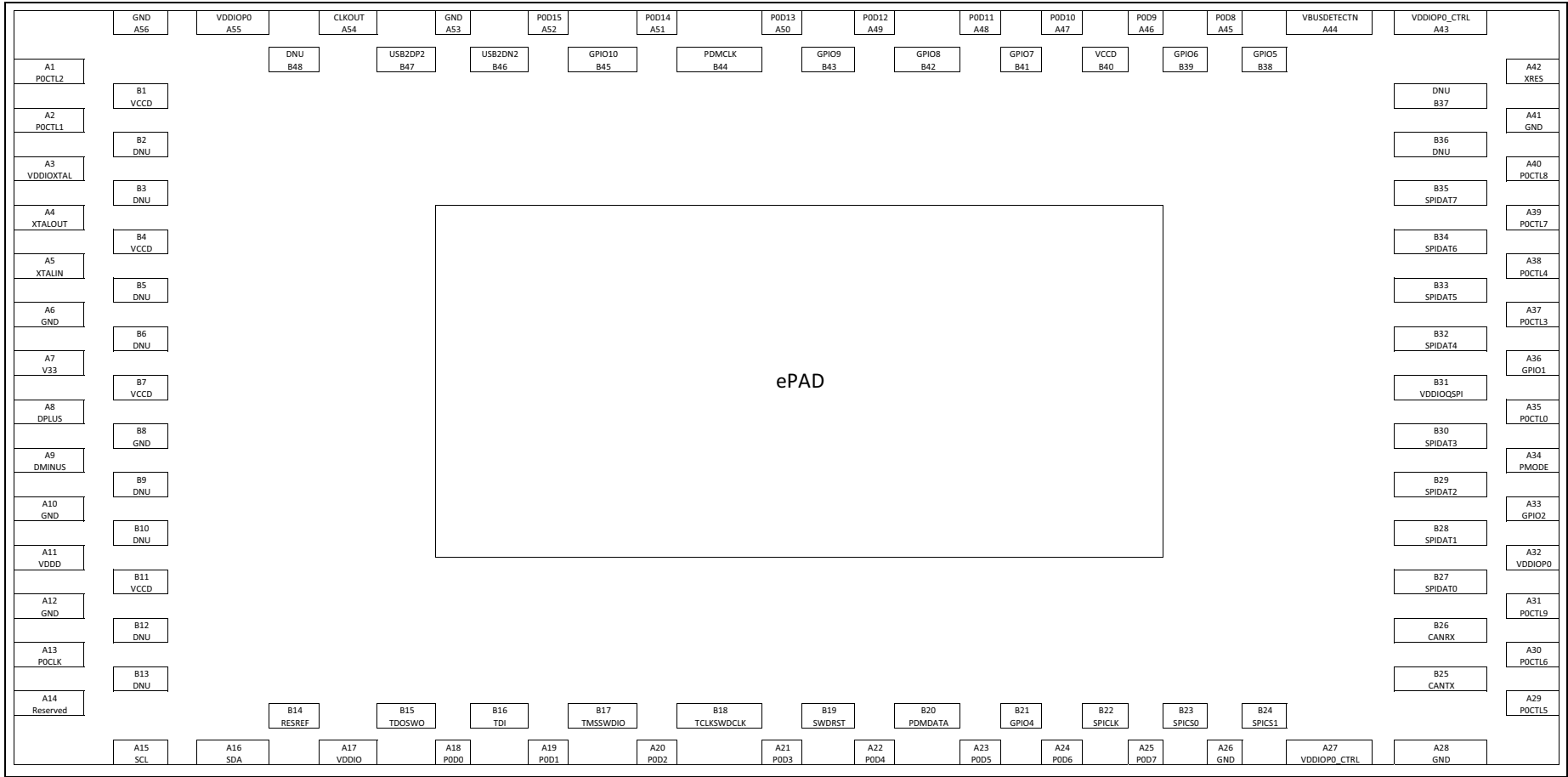


Figure 4 Product ball map (top view)

Pinouts

3.1 Pin definitions

Table 8 Pin definitions

SL#	Pin#	CY7C68315-BF104AXI	CYUSB2316-BF104AXI	CYUSB2317-BF104AXI /CYUSB2318-BF104AXI	Port/ pin	Type of signal	Associated supply	Description
		Pin name	Pin name	Pin name				
1	A1	P0CTL2	P0CTL2	P0CTL2	–	LVC MOS Control	VDDIOP0_CTRL	–
2	A2	P0CTL1	P0CTL1	P0CTL1	–	LVC MOS Control	VDDIOP0_CTRL	–
3	A3	VDDIOXTAL	VDDIOXTAL	VDDIOXTAL	–	Crystal Power	VDDIOXTAL	–
4	A4	XTALOUT	XTALOUT	XTALOUT	–	Crystal Output	VDDIOXTAL	XTALOUT / OSC IN
5	A5	XTALIN	XTALIN	XTALIN	–	Crystal Input	VDDIOXTAL	XTALIN / OSC IN
6	A6	GND	GND	GND	–	Ground	-	–
7	A7	V33	V33	V33	–	Power	V33	–
8	A8	DPLUS	DPLUS	DPLUS	–	USB	V33	–
9	A9	DMINUS	DMINUS	DMINUS	–	USB	V33	–
10	A10	GND	GND	GND	–	Ground	-	–
11	A11	VDDD	VDDD	VDDD	–	Power	VDDD	–
12	A12	GND	GND	GND	–	Ground	-	–
13	A13	P0CLK	P0CLK	P0CLK	–	LVC MOS Clock	VDDIOP0	–
14	A14	Reserved	Reserved	Reserved	–	Input	–	Connect to GND
15	A15	SCL	SCL	SCL	P10.0	I2C Clock	VDDIO	–
16	A16	SDA	SDA	SDA	P10.1	I2C Data	VDDIO	–
17	A17	VDDIO	VDDIO	VDDIO	–	Power	VDDIO	–
18	A18	P0D0	P0D0	P0D0	–	LVC MOS Data	VDDIOP0	–
19	A19	P0D1	P0D1	P0D1	–	LVC MOS Data	VDDIOP0	–
20	A20	P0D2	P0D2	P0D2	–	LVC MOS Data	VDDIOP0	–

Pinouts

Table 8 Pin definitions

SL#	Pin#	CY7C68315-BF104AXI	CYUSB2316-BF104AXI	CYUSB2317-BF104AXI /CYUSB2318-BF104AXI	Port/ pin	Type of signal	Associated supply	Description
		Pin name	Pin name	Pin name				
21	A21	P0D3	P0D3	P0D3	–	LVC MOS Data	VDDIOP0	–
22	A22	P0D4	P0D4	P0D4	–	LVC MOS Data	VDDIOP0	–
23	A23	P0D5	P0D5	P0D5	–	LVC MOS Data	VDDIOP0	–
24	A24	P0D6	P0D6	P0D6	–	LVC MOS Data	VDDIOP0	–
25	A25	P0D7	P0D7	P0D7	–	LVC MOS Data	VDDIOP0	–
26	A26	GND	GND	GND	–	Ground	–	–
27	A27	VDDIOP0_CTRL	VDDIOP0_CTRL	VDDIOP0_CTRL	–	Power	VDDIOP0_CTRL	–
28	A28	GND	GND	GND	–	Ground	–	–
29	A29	P0CTL5	P0CTL5	P0CTL5	–	LVC MOS Control	VDDIOP0_CTRL	–
30	A30	P0CTL6	P0CTL6	P0CTL6	–	LVC MOS Control	VDDIOP0_CTRL	–
31	A31	P0CTL9	P0CTL9	P0CTL9	–	LVC MOS Control	VDDIOP0_CTRL	–
32	A32	VDDIOP0	VDDIOP0	VDDIOP0	–	Power	VDDIOP0	–
33	A33	GPIO2	GPIO2	GPIO2	P4.4	GPIO	VDDIO	Alternate mode: SPI-CS0 (SCB5)
34	A34	PMODE	PMODE	PMODE	P13.0	GPIO	VDDIO	Alternate mode: SPI-MOSI (SCB5)
35	A35	P0CTL0	P0CTL0	P0CTL0	–	LVC MOS Control	VDDIOP0_CTRL	–
36	A36	GPIO1	GPIO1	GPIO1	P4.3	GPIO	VDDIO	Alternate mode: SPICLK (SCB5)
37	A37	P0CTL3	P0CTL3	P0CTL3	–	LVC MOS Control	VDDIOP0_CTRL	–
38	A38	P0CTL4	P0CTL4	P0CTL4	–	LVC MOS Control	VDDIOP0_CTRL	–
39	A39	P0CTL7	P0CTL7	P0CTL7	–	LVC MOS Control	VDDIOP0_CTRL	–
40	A40	P0CTL8	P0CTL8	P0CTL8	–	LVC MOS Control	VDDIOP0_CTRL	–

Pinouts

Table 8 Pin definitions

SL#	Pin#	CY7C68315-BF104AXI	CYUSB2316-BF104AXI	CYUSB2317-BF104AXI /CYUSB2318-BF104AXI	Port/ pin	Type of signal	Associated supply	Description
		Pin name	Pin name	Pin name				
41	A41	GND	GND	GND	–	Ground	–	–
42	A42	XRES	XRES	XRES	–	GPIO	VDDD	–
43	A43	VDDIOP0_CTRL	VDDIOP0_CTRL	VDDIOP0_CTRL	–	Power	VDDIOP0_CTRL	–
44	A44	VBUSDETECTN	VBUSDETECTN	VBUSDETECTN	P4.0	GPIO	VDDIO	–
45	A45	P0D8	P0D8	P0D8	–	LVC MOS Data	VDDIOP0	–
46	A46	P0D9	P0D9	P0D9	–	LVC MOS Data	VDDIOP0	–
47	A47	P0D10	P0D10	P0D10	–	LVC MOS Data	VDDIOP0	–
48	A48	P0D11	P0D11	P0D11	–	LVC MOS Data	VDDIOP0	–
49	A49	P0D12	P0D12	P0D12	–	LVC MOS Data	VDDIOP0	–
50	A50	P0D13	P0D13	P0D13	–	LVC MOS Data	VDDIOP0	–
51	A51	P0D14	P0D14	P0D14	–	LVC MOS Data	VDDIOP0	–
52	A52	P0D15	P0D15	P0D15	–	LVC MOS Data	VDDIOP0	–
53	A53	GND	GND	GND	–	Ground	–	–
54	A54	CLKOUT	CLKOUT	CLKOUT	–	CLKOUT	VDDIOP0	–
55	A55	VDDIOP0	VDDIOP0	VDDIOP0	–	Power	VDDIOP0	–
56	A56	GND	GND	GND	–	Ground	–	–
57	ePAD	GND	GND	GND	–	Ground	–	–
58	B1	DNU	VCCD	VCCD1	–	Power	VCCD	Short with VCCD in A0 silicon. DNU in B0 silicon.
59	B2	DNU	DNU	DNU	–	NA	–	–
60	B3	DNU	DNU	DNU	–	NA	–	–

Pinouts

Table 8 Pin definitions

SL#	Pin#	CY7C68315-BF104AXI	CYUSB2316-BF104AXI	CYUSB2317-BF104AXI /CYUSB2318-BF104AXI	Port/ pin	Type of signal	Associated supply	Description
		Pin name	Pin name	Pin name				
61	B4	DNU	VCCD	VCCD2	–	Power	VCCD	Short with VCCD in A0 silicon. DNU in B0 silicon.
62	B5	DNU	DNU	DNU	–	NA	–	–
63	B6	DNU	DNU	DNU	–	NA	–	–
64	B7	DNU	VCCD	VCCD3	–	Power	VCCD	Short with VCCD in A0 silicon. DNU in B0 silicon
65	B8	DNU	GND	GND	–	Ground	–	–
66	B9	DNU	DNU	DNU	–	NA	–	–
67	B10	DNU	DNU	DNU	–	NA	–	–
68	B11	DNU	VCCD	VCCD4	–	Power	VCCD	Short with VCCD in A0 silicon. DNU in B0 silicon
69	B12	DNU	DNU	DNU	–	NA	–	–
70	B13	DNU	DNU	DNU	–	NA	–	–
71	B14	RESREF	RESREF	RESREF	–	Input	–	Connect a 6.04 KΩ ± 1% resistor to GND
72	B15	DNU	TDOSWO	TDOSWO	P11.0	GPIO	VDDIO	–
73	B16	DNU	TDI	TDI	P11.1	GPIO	VDDIO	–
74	B17	DNU	TMSSWDIO	TMSSWDIO	P11.2	GPIO	VDDIO	–
75	B18	DNU	TCLKSWDCLK	TCLKSWDCLK	P11.3	GPIO	VDDIO	–
76	B19	DNU	SWDRST	SWDRST	P11.4	GPIO	VDDIO	–
77	B20	DNU	PDMDATA	PDMDATA	P9.1	GPIO	VDDIO	Alternate mode: TCPWM4_N

Pinouts

Table 8 Pin definitions

SL#	Pin#	CY7C68315-BF104AXI	CYUSB2316-BF104AXI	CYUSB2317-BF104AXI /CYUSB2318-BF104AXI	Port/ pin	Type of signal	Associated supply	Description
		Pin name	Pin name	Pin name				
78	B21	DNU	GPIO4	GPIO4	P13.1	GPIO	VDDIO	Alternate mode: SPI-MISO (SCB5)
79	B22	DNU	DNU	SPICLK	P6.0	GPIO	VDDIOQSPI	–
80	B23	DNU	DNU	SPICS0	P6.1	GPIO	VDDIOQSPI	–
81	B24	DNU	DNU	SPICS1	P6.2	GPIO	VDDIOQSPI	–
82	B25	DNU	CANTX	CANTX	P8.5	GPIO	VDDIOQSPI	Alternate mode: UART-TX TCPWM2_N
83	B26	DNU	CANRX	CANRX	P8.4	GPIO	VDDIOQSPI	Alternate mode: UART-RX TCPWM2_P
84	B27	DNU	DNU	SPIDAT0	P7.0	GPIO	VDDIOQSPI	–
85	B28	DNU	DNU	SPIDAT1	P7.1	GPIO	VDDIOQSPI	–
86	B29	DNU	DNU	SPIDAT2	P7.2	GPIO	VDDIOQSPI	–
87	B30	DNU	DNU	SPIDAT3	P7.3	GPIO	VDDIOQSPI	–
88	B31	DNU	DNU	VDDIOQSPI	–	Power	VDDIOQSPI	–
89	B32	DNU	DNU	SPIDAT4	P7.4	GPIO	VDDIOQSPI	–
90	B33	DNU	DNU	SPIDAT5	P7.5	GPIO	VDDIOQSPI	–
91	B34	DNU	DNU	SPIDAT6	P7.6	GPIO	VDDIOQSPI	–
92	B35	DNU	DNU	SPIDAT7	P7.7	GPIO	VDDIOQSPI	–
93	B36	DNU	DNU	DNU	P1.0	–	VDDIO	–
94	B37	DNU	DNU	DNU	P1.1	–	VDDIO	–
95	B38	DNU	GPIO5	GPIO5	–	GPIO	VDDIOP0	–
96	B39	DNU	GPIO6	GPIO6	–	GPIO	VDDIOP0	–

Pinouts

Table 8 Pin definitions

SL#	Pin#	CY7C68315-BF104AXI	CYUSB2316-BF104AXI	CYUSB2317-BF104AXI /CYUSB2318-BF104AXI	Port/ pin	Type of signal	Associated supply	Description
		Pin name	Pin name	Pin name				
97	B40	VCCD	VCCD	VCCD	–	Power	VCCD	Connect 4.7 μ F $\pm$ 20% from this pin to GND
98	B41	DNU	GPIO7	GPIO7	–	GPIO	VDDIOP0	–
99	B42	DNU	GPIO8	GPIO8	–	GPIO	VDDIOP0	–
100	B43	DNU	GPIO9	GPIO9	–	GPIO	VDDIOP0	–
101	B44	DNU	PDMCLK	PDMCLK	P9.0	GPIO	VDDIO	Alternate mode: TCPWM4_P
102	B45	DNU	GPIO10	GPIO10	–	GPIO	VDDIOP0	–
103	B46	DNU	USB2DN2	USB2DN2	–	USB	V33	–
104	B47	DNU	USB2DP2	USB2DP2	–	USB	V33	–
105	B48	DNU	DNU	DNU	–	NA	NA	–

Pinouts

Table 9 Peripheral and other signals

Pin#	Category	Signal name		Type of signal	Signal description	Associated supply
		Function after reset	Alternate functions			
B22	SPI/QSPI/ Octal SPI	P6.0	SPICLK (SMIF) UART-RX1 (SCB6) I2C-SCL1 (SCB6) SPICLK (SCB3)	GPIO	–	VDDIOQSPI
B23		P6.1	SPICS0 (SMIF) UART-TX1 (SCB6) I2C-SDA1 (SCB6) SPICS0 (SCB3)	GPIO	–	VDDIOQSPI
B24		P6.2	SPICS1 (SMIF) SPI-MOSI (SCB3)	GPIO	–	VDDIOQSPI
B27		P7.0	SPIDAT0 (SMIF) UART-RX1 (SCB0) I2C-SCL (SCB0) SPICLK (SCB2)	GPIO	–	VDDIOQSPI
B28		P7.1	SPIDAT1 (SMIF) UART-TX1 (SCB0) I2C-SDA (SCB0) SPICS0 (SCB2)	GPIO	–	VDDIOQSPI
B29		P7.2	SPIDAT2 (SMIF) UART-CTS0 (SCB0) SPI-MOSI (SCB2)	GPIO	–	VDDIOQSPI
B30		P7.3	SPIDAT3 (SMIF) UART-CTS0 (SCB3) UART-RTS0 (SCB0) SPI-MISO (SCB2)	GPIO	–	VDDIOQSPI
B32		P7.4	SPIDAT4 (SMIF) UART-CTS0 (SCB2)	GPIO	–	VDDIOQSPI
B33		P7.5	SPIDAT5 (SMIF)	GPIO	–	VDDIOQSPI
B34		P7.6	SPIDAT6 (SMIF) UART-RX1 (SCB1) I2C-SDA1 (SCB1)	GPIO	–	VDDIOQSPI
B35		P7.7	SPIDAT7 (SMIF) UART-TX1 (SCB1) I2C-SCL1 (SCB1)	GPIO	–	VDDIOQSPI
A36	User GPIOs	P4.3	GPIO1 UART-RX0 I2C-SCL0 (SCB5) SPICLK (SCB5)	GPIO	–	VDDIO
A33		P4.4	GPIO2 UART-TX0 I2C-SDA0 (SCB5) SPI-CS0 (SCB5) CAN1-RX	GPIO	–	VDDIO
A34		P13.0	PMODE SPI-MOSI (SCB5) CAN1-TX	GPIO	–	VDDIO
A44		P13.1	GPIO4 SPI-MISO (SCB5)	GPIO	–	VDDIO

Pinouts

Table 9 Peripheral and other signals (continued)

Pin#	Category	Signal name		Type of signal	Signal description	Associated supply
		Function after reset	Alternate functions			
A15	I2C	P10.0	I2C-SCL0 (SCB0) UART-RX0 (SCB0)	OVT GPIO	I2C Clock (SCL)	VDDIO
A16		P10.1	I2C-SDA0 (SCB0) UART-TX0 (SCB0)	OVT GPIO	I2C Data (SDA)	VDDIO
B19	SWD/JTAG	P11.4	SWDRST I2C-SCL1 (SCB5) CAN3-RX	GPIO	–	VDDIO
B18		P11.3	TCLKSWDCLK PDM3-DATA1 UART-CTS1 (SCB4) SPI-MISO (SCB4)	GPIO	–	VDDIO
B17		P11.2	TMSSWDIO PDM3-CLK1 UART-RTS1 (SCB4) SPI-MOSI (SCB4)	GPIO	–	VDDIO
B16		P11.1	TDI PDM3-DATA0 UART-RX1 (SCB4) I2C-SDA1 (SCB4) SPIC0 (SCB4)	GPIO	–	VDDIO
B15		P11.0	TDOSWO PDM3-CLK0 UART-TX1 (SCB4) I2C-SCL1 (SCB4) SPICLK (SCB4)	GPIO	–	VDDIO
B25	CAN FD	P8.5	CAN0-TX PDM0-DATA1 UART-TX0 (SCB4) I2C-SDA0 (SCB4) TCPWM2_N SPIC0 (SCB4)	GPIO	–	VDDIO
B26		P8.4	CAN0-RX PDM0-CLK1 UART-RX0 (SCB4) I2C-SCL0 (SCB4) TCPWM2_P SPICLK (SCB4)	GPIO	–	VDDIO

Points to be taken care while migrating from
FX2LP to FX2G3

4 Points to be taken care while migrating from FX2LP to FX2G3

- The 104 pins of the FX2G3 are arranged in two rows. The outer ring is pin compatible to 56-pin QFN package of FX2LP. However, two additional pins namely RESREF and VCCD that are present in the inner ring need to be used to connect external passives as described in [Table 8](#).
- In order to enable existing FX2LP customers to use FX2G3 as drop in the existing designs using SSOP or TQFP packages of FX2LP, the KIT_FX2G3_104LGA will come with FX2G3 based adapter boards. These boards will allow users to replace FX2LP SSOP and TQFP packages on their boards, enable to migrate the firmware, and plan for hardware change.
- FX2LP is 8051 controller-based architecture whereas FX2G3 is ARM M0+, M4 core-based architecture. Hence, the firmware needs to be rewritten for FX2G3.

5 Power supply consideration

Figure 5 shows the power system requirements for power pins for all supported packages.

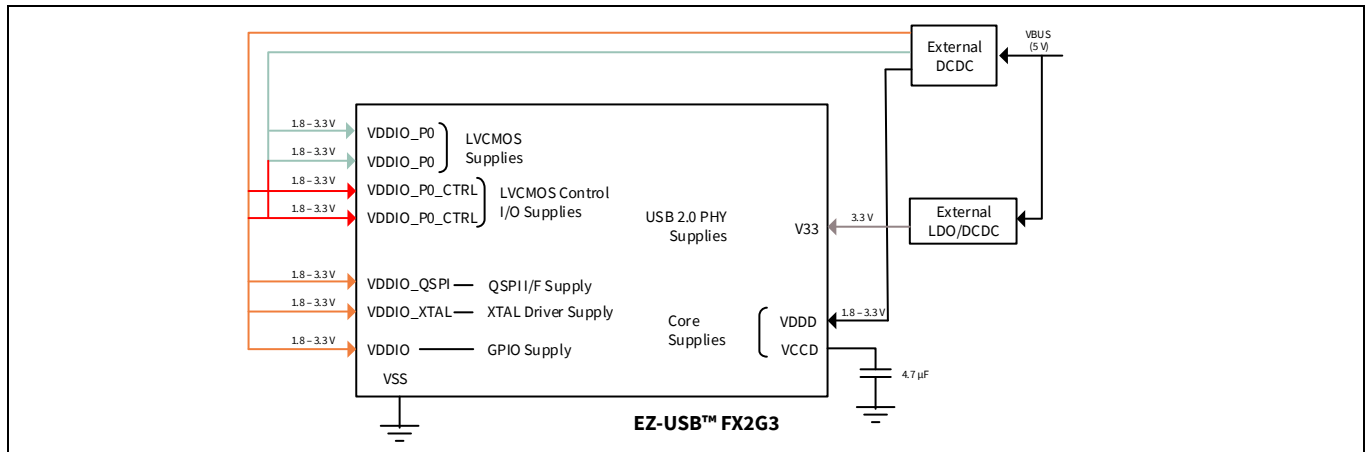


Figure 5 Power system requirement block diagram

5.1 Core power domains

- VDDD is the main digital supply input. It provides the inputs for the internal regulators.
- VCCD is the internal core logic supply for the 104LGA part and needs to be decoupled externally
- VDDIO_QSPI is the power supply for the QSPI port
- VDDIO_XTAL is the power supply for crystal
- VDDIO is the GPIO power supply
- VSS is the main ground for the entire chip

5.2 USB 2.0 PHY power supplies

- V33 is the 3.3V power supply for the USB2 HS PHY and USB2 FS PHY.
- VDDD is the power supply for the USB2 HS PHY TX IO circuitry.

5.3 LVC MOS port power supplies

- VDDIO_P0 are the power supplies for LVC MOS IOs
- VDDIO_P0_CTRL are the power supplies for the LVC MOS control IOs

6 Electrical specifications

All specifications are valid for $-40^{\circ}\text{C} \leq \text{TA} \leq 85^{\circ}\text{C}$ and for 1.71 V to 3.6 V except where noted.

6.1 Absolute maximum ratings

Table 10 Absolute maximum ratings

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID-ABS1	VDDD_ABS	Analog or digital supply relative to V_{SS}	-0.3	-	4.02	V	-
SID-ABS3	V33_ABS	Max voltage on V33 supply pin	-0.3	-	4.02	V	-
SID-ABS4	VDDD_ABS	Max voltage on VDDD supply pin	-0.3	-	4.02	V	-
SID-ABS5	VDDIO_ABS	Max voltage on VDDIO, VDDIO_* supply pins	-0.3	-	4.02	V	-
SID-ABS6	VGPIIO_OVT_ABS	OVT GPIO voltage	-0.3	-	4.02	V	-
SID-ABS7	IGPIIO_ABS_DS0	Current per GPIO at DS = 0	-7.5	-	7.5	mA	-
SID-ABS8	IGPIIO_ABS_DS1	Current per GPIO at DS = 1	-5.6	-	5.6	mA	-
SID-ABS9	IGPIIO_ABS_DS2	Current per GPIO at DS = 2	-3.8	-	3.8	mA	-
SID-ABS10	IGPIIO_ABS_DS3	Current per GPIO at DS = 3	-1.9	-	1.9	mA	-
SID-ABS11	ESD_HBM	Electrostatic discharge human body model (ESD HBM) JEDEC JS-001-2017	2000	-	-	V	-
SID-ABS12	ESD_CDM	Electrostatic discharge charged device model (ESD CDM) JS-002-2018	500	-	-	V	-
SID-ABS13	LU	Pin current for latch-up free operation	-100	-	100	mA	-
SID-ABS14	TJ	Junction temperature	-	-	125	$^{\circ}\text{C}$	-
SID-ABS15	TSTG	Storage temperature	-65	-	150	$^{\circ}\text{C}$	-
SID-ABS16	TSIT	Lead temperature (soldering 4 s)	-	-	260	$^{\circ}\text{C}$	-
SID-ABS17	TA	Ambient temperature	-40	-	85	$^{\circ}\text{C}$	-

Table 11 Pin-based absolute maximum ratings

Category	Pin name	Absolute min. (V)	Absolute max. (V)
LVC MOS	P[0][C,D,CLK,CTL]	-0.5	4.0
USB HS	USBHS[DP,DN]	-0.5	4.0
USB FS	USBFS[DP,DN]	-0.5	4.0
CLOCK	XTAL[IN,OUT]	-0.5	4.0
RESET	XRES	-0.5	4.0
SPI / QSPI / Octal SPI	P[6,7]	-0.5	4.0
UART, PDM, GPIO, SWD, JTAG, CAN FD, USB Ctrl	P[1,4,6,7,8,9,11]	-0.5	4.0

Electrical specifications

6.2 DC specifications**Table 12 DC operating specifications**

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID-DC1	VDDD	Main analog / digital supply voltage	1.62	1.8 or 3.3	3.63	V	–
SID-DC2	VDDD_EFUSE	VDDD supply when programming eFuse	2.38	2.5	2.62	V	eFuse programming voltage
SID-DC4	V33	Power supply voltage for 3.3 V supply	2.97	3.3	3.63	V	–
SID-DC6	VDDIO_1P8	Power supply voltage for IO in the 1.8 V range	1.62	1.8	1.95	V	–
SID-DC7	VDDIO_2P5	Power supply voltage for IO in the 2.5 V range	2.25	2.5	2.75	V	–
SID-DC8	VDDIO_3P3	Power supply voltage for IO in the 3.3 V range	2.97	3.3	3.63	V	–
SID-DC9	VCCD	Core logic supply output	–	1.1	–	V	–
SID-DC10	CDEC1	Decoupling capacitor on VCCD	1.88	4.7	6.58	μF	Capacitor must have initial tolerance range of 4.7 μF ±20%

6.3 Power specifications**Table 13 Power specifications**

VDDD = 1.8 V (typ), 1.98 V (max)

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID-PWR16	FX2G3_ACTIVE	USB 2.0 mode, LVCMOS @ 48 MHz SDR	–	129.45	199.13	mW	Default setting (Amp setting = 4, Pre-emphasis setting = 4, Pre-emphasis duration = 2)
SID-PWR17	FX2G3_SUSPEND	U3 Suspend (Deep Sleep), USB 2.0 in U3, LVCMOS port shutdown	–	0.85	–	mW	–
SID-PWR14	IDLE	Deep Sleep mode	–	160	–	μW	–
SID-PWR15	Power down	XRES(RESET_N) asserted	–	80	–	μW	–

Electrical specifications

Table 14 Power specifications**Note** VDDD = 3.3 V (typ), 3.63 V (max)

Spec ID#	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID-PWR16A	FX2G3_ACTIVE	USB 2.0 mode, LVCMOS @ 48 MHz SDR	–	222.32	360.84	mW	Default setting (Amp setting = 4, Pre-emphasis setting = 4, Pre-emphasis duration = 2)
SID-PWR17A	FX2G3_SUSPEND	U3 Suspend (Deep sleep), USB 2.0 in U3, LVCMOS Port shutdown	–	0.97	–	mW	–
SID-PWR14A	IDLE	Deep Sleep mode	–	300	–	μW	–
SID-PWR15A	Power Down	XRES(RESET_N) asserted	–	220	–	μW	–

6.4 RESETB requirements**Table 15 RESETB requirements**

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_RESETB1	Vih_RESETB	Input voltage high threshold on RESETB pin	65	–	–	%	CMOS Input, % of VDDD supply
SID_RESETB2	ViL_RESETB	Input voltage low threshold on RESETB pin	–	–	35	%	CMOS Input, % of VDDD supply
SID_RESETB3	Cin_RESETB	Input capacitance on RESETB pin	–	–	7	pF	–
SID_RESETB4	TRESETB	External reset pulse width	1	–	–	μs	–40°C to +85°C TA, all VDDIO
SID_RESETB5	TRESETB_GF	External reset glitch filter period	–	–	20	ns	–40°C to +85°C TA, all VDDIO
SID_RESETB6	RPD_RESETB	Pull-down resistance of RESETB pin	60	85	110	kΩ	–
SID_RESETB7	RESETB_TREADY	Time from RESETB deassertion to device being ready for USB traffic	–	–	32	ms	CMOS Input, % of VDDD supply

Electrical specifications

6.5 Standard GPIO DC specification

Table 16 Standard GPIO DC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_GIO1	RPU	Pull-up resistor value	3.5	–	8.5	kΩ	–
SID_GIO2	RPD	Pull-down resistor value	3.5	–	8.5	kΩ	–
SID_GIO3	IIL	Input leakage current (absolute value)	-1	–	1	μA	–
SID_GIO4	CPIN	Maximum pin capacitance	–	–	9.5	pF	–
SID_GIO5	Vih_gpio	Input voltage HIGH level	0.65 * VDDIO	–		V	–
SID_GIO6	Vil_gpio	Input voltage LOW level	–	–	0.35 * VDDIO	V	–
SID_GIO7	FGPIO_OUT0_v1 p8	Output frequency at 1.8 V, drive_sel<1:0>= 00, slow=0	–	–	30	MHz	Load 25 pF, 90%/10%
SID_GIO8	FGPIO_OUT1_v1 p8	Output frequency at 1.8 V, drive_sel<1:0>= 01, slow=0	–	–	15	MHz	Load 25 pF, 90%/10%
SID_GIO9	FGPIO_OUT2_v1 p8	Output frequency at 1.8 V, drive_sel<1:0>= 10, slow=0	–	–	7.5	MHz	Load 25 pF, 90%/10%
SID_GIO10	FGPIO_OUT3_v1 p8	Output frequency at 1.8 V, drive_sel<1:0>= 11, slow=0	–	–	2.5	MHz	Load 25 pF, 90%/10%
SID_GIO11	FGPIO_OUT0_v2 p5	Output frequency at 2.5 V, drive_sel<1:0>= 00, slow=0	–	–	40	MHz	Load 25 pF, 90%/10%
SID_GIO12	FGPIO_OUT1_v2 p5	Output frequency at 2.5 V, drive_sel<1:0>= 01, slow=0	–	–	20	MHz	Load 25 pF, 90%/10%
SID_GIO13	FGPIO_OUT2_v2 p5	Output frequency at 2.5 V, drive_sel<1:0>= 10, slow=0	–	–	12.5	MHz	Load 25 pF, 90%/10%
SID_GIO14	FGPIO_OUT3_v2 p5	Output frequency at 2.5 V, drive_sel<1:0>= 11, slow=0	–	–	5	MHz	Load 25 pF, 90%/10%
SID_GIO15	FGPIO_OUT0_v3 p3	Output frequency at 3.3 V, drive_sel<1:0>= 00, slow=0	–	–	50	MHz	Load 25 pF, 90%/10%
SID_GIO16	FGPIO_OUT1_v3 p3	Output frequency at 3.3 V, drive_sel<1:0>= 01, slow=0	–	–	25	MHz	Load 25 pF, 90%/10%
SID_GIO17	FGPIO_OUT2_v3 p3	Output frequency at 3.3 V, drive_sel<1:0>= 10, slow=0	–	–	15	MHz	Load 25 pF, 90%/10%
SID_GIO18	FGPIO_OUT3_v3 p3	Output frequency at 3.3 V, drive_sel<1:0>= 11, slow=0	–	–	7.5	MHz	Load 25 pF, 90%/10%
SID_GIO19	FGPIO_IN	GPIO input operating frequency	–	–	50	MHz	–
SID_GIO20	IOH_v1p8_0	1.8 V pull-up current, DS=0	-2.12	-4	-5.88	mA	–
SID_GIO21	IOH_v1p8_1	1.8 V pull-up current, DS=1	-1.06	-2	-2.94	mA	–
SID_GIO22	IOH_v1p8_2	1.8 V pull-up current, DS=2	-0.8	-1.5	-2.2	mA	–
SID_GIO23	IOH_v1p8_3	1.8 V pull-up current, DS=3	-0.27	-0.5	-0.73	mA	–
SID_GIO24	IOH_v2p5_0	2.5 V pull-up current, DS=0	-3.18	-6	-8.82	mA	–
SID_GIO25	IOH_v2p5_1	2.5 V pull-up current, DS=1	-1.59	-3	-4.41	mA	–
SID_GIO26	IOH_v2p5_2	2.5 V pull-up current, DS=2	-1.06	-2	-2.94	mA	–
SID_GIO27	IOH_v2p5_3	2.5 V pull-up current, DS=3	-0.4	-0.75	-1.1	mA	–

Electrical specifications

Table 16 **Standard GPIO DC specifications** (continued)

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_GIO28	IOH_v3p3_0	3.3 V pull-up current, DS=0	-4.24	-8	-11.76	mA	-
SID_GIO29	IOH_v3p3_1	3.3 V pull-up current, DS=1	-2.12	-4	-5.88	mA	-
SID_GIO30	IOH_v3p3_2	3.3 V pull-up current, DS=2	-1.59	-3	-4.41	mA	-
SID_GIO31	IOH_v3p3_3	3.3 V pull-up current, DS=3	-0.53	-1	-1.47	mA	-
SID_GIO32	IOL_v1p8_0	1.8 V pull-down current, DS=0	2.12	4	5.88	mA	-
SID_GIO33	IOL_v1p8_1	1.8 V pull-down current, DS=1	1.06	2	2.94	mA	-
SID_GIO34	IOL_v1p8_2	1.8 V pull-down current, DS=2	0.8	1.5	2.2	mA	-
SID_GIO35	IOL_v1p8_3	1.8 V pull-down current, DS=3	0.27	0.5	0.73	mA	-
SID_GIO36	IOL_v2p5_0	2.5 V pull-down current, DS=0	3.18	6	8.82	mA	-
SID_GIO37	IOL_v2p5_1	2.5 V pull-down current, DS=1	1.59	3	4.41	mA	-
SID_GIO38	IOL_v2p5_2	2.5 V pull-down current, DS=2	1.06	2	2.94	mA	-
SID_GIO39	IOL_v2p5_3	2.5 V pull-down current, DS=3	0.4	0.75	1.1	mA	-
SID_GIO40	IOL_v3p3_0	3.3 V pull-down current, DS=0	4.24	8	11.76	mA	-
SID_GIO41	IOL_v3p3_1	3.3 V pull-down current, DS=1	2.12	4	5.88	mA	-
SID_GIO42	IOL_v3p3_2	3.3 V pull-down current, DS=2	1.59	3	4.41	mA	-
SID_GIO43	IOL_v3p3_3	3.3 V pull-down current, DS=3	0.53	1	1.47	mA	-
SID_GIO44	TRF_v1p8	Output rise/fall time at 1.8 V, drive_sel<1:0>= 11, slow=0	-	-	50	ns	-
SID_GIO45	TRF_v3p3	Output rise/fall time at 3.3 V, drive_sel<1:0>= 00, slow=0	-	-	7	ns	-
SID_GIO46	IIHS	IO pin current when Vpad is ramped up from 0 to 3.63 V and VDDIO = 0, T = 25°C	-	-	1	μA	-

Electrical specifications

6.6 SWD and trace interface specification (applicable only for 104LGA part)

Table 17 SWD and trace interface specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_SWD1	F_SWCLK	SWD input clock frequency	–	–	25	MHz	1.62 V ≤ VDDIO ≤ 3.63 V
SID_SWD2	T_SWDI_SETUP	T = 1/f SWDCLK	0.25 × T	–	–	ns	–
SID_SWD3	T_SWDI_HOLD	T = 1/f SWDCLK	0.25 × T	–	–	ns	–
SID_SWD4	T_SWDO_VALID	T = 1/f SWDCLK	–	–	0.50 × T	ns	–
SID_SWD5	T_SWDO_HOLD	T = 1/f SWDCLK	1	–	–	ns	–
SID_SWD6	F_TRCLK_LP1	With trace data setup/hold times of 2/1 ns respectively	–	–	48	MHz	–
SID_SWD7	F_TRCLK_LP2	With trace data setup/hold times of 3/2 ns respectively	–	–	48	MHz	–
SID_SWD8	SWD_INPUT_TRANSITION_TIME	SWD input transition time	–	–	5	ns	–
SID_SWD9	TRACE_CLKL	Trace clock LOW Period	2	–	–	ns	–
SID_SWD10	TRACE_CLKH	Trace clock HIGH Period	2	–	–	ns	–
SID_SWD11	TRACE_CLKL	Trace clock minimum pulse-width LOW	8	–	–	ns	–
SID_SWD12	TRACE_CLKH	Trace clock minimum pulse-width HIGH	8	–	–	ns	–

6.7 Precise power-on reset (POR) specification

Table 18 POR with brownout DC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_PORBOD1	VFALLPPOR	BOD trip voltage in Active and Sleep modes VDDD	1.5	–	–	V	BOD reset guaranteed for levels < 1.5 V
SID_PORBOD2	VFALLDPSLP	BOD trip voltage in Deep Sleep VDDD	1.5	–	–	V	–
SID_PORBOD3	VDDRAMP	Maximum power supply ramp rate (any supply)	–	–	100	mV/μs	Active mode

Table 19 POR with brownout AC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_PORBOD4	VDDRAMP_DS	Maximum power supply ramp rate (any supply) in Deep Sleep	–	–	12.5	mV/μs	BOD operation guaranteed

Electrical specifications

6.8 XTAL / clock specification**Table 20 XTAL / clock specifications**

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_ECO1	EXT_CRY_RES_FREQ	External crystal resonator frequency	–	24	–	MHz	Resonator reference clock frequency. Fundamental oscillation mode
SID_ECO2	EXT_CRY_RES_FREQ_TOL	External crystal resonator frequency tolerance	–30	–	30	ppm	Resonator reference clock accuracy
SID_ECO3	EXT_CRY_RES_LOAD	External crystal resonator load capacitance	–	9.5	–	pF	PCB load
SID_ECO4	EXT_CRY_RES_CSHUNT	External crystal resonator shunt capacitance	–	0.4	1	pF	No ground connection
SID_ECO5	EXT_CRY_RESCOTIONAL	External crystal resonator motional capacitance	–	–	3	pF	No ground connection
SID_ECO6	EXT_CRY_RES_ESR	External crystal resonator ESR	–	–	60	Ω	Equivalent sales resistance
SID_ECO7	EXT_CRY_RES_PDRIVE	Drive level for external crystal oscillator	–	50	200	μW	Resonator drive level
SID_ECO8	EXT_CRY_RES_START	External crystal resonator start-up time	–	–	1.6	ms	Maximum start-up time required by resonator to meet the USB 2.0 timing requirements.

6.9 Single-ended LVCMOS reference clock specification

Table 21 Single-ended LVCMOS reference clock input specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_REF1	LVCNOS_ REFCLK_FREQ	LVCNOS reference clock frequency	–	24	–	MHz	–
SID_REF2	LVCNOS_ REFCLK_ FREQ_TOL	LVCNOS reference clock frequency tolerance	–30	–	30	ppm	–
SID_REF3	LVCNOS_ REFCLK_ FREQ_DUTY	LVCNOS reference clock frequency duty cycle	40	–	60	%	–
SID_REF4	LVCNOS_ REFCLK_VIH	Input voltage HIGH threshold for LVCNOS reference clock	0.65*VDDIO	–	–	V	–
SID_REF5	LVCNOS_ REFCLK_VIL	Input voltage LOW threshold for LVCNOS reference clock	–	–	0.35*VDDIO	V	–
SID_REF6	LVCNOS_ REFCLK_CIN	Input capacitance for LVCNOS reference clock	–	–	7	pF	–

Table 22 Single-ended CMOS reference clock input phase noise requirement

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_PH1	IN_PH_NOISE_100	Maximum input phase noise @ 100-Hz offset	–	–	–75	dBc/Hz	–
SID_PH2	IN_PH_NOISE_1K	Maximum input phase noise @ 1-kHz offset	–	–	–104	dBc/Hz	–
SID_PH3	IN_PH_NOISE_10K	Maximum input phase noise @ 10-kHz offset	–	–	–120	dBc/Hz	–
SID_PH4	IN_PH_NOISE_100K	Maximum input phase noise@ 100-kHz offset	–	–	–128	dBc/Hz	–
SID_PH5	IN_PH_NOISE_1M	Maximum input phase noise @ 1-MHz offset	–	–	–130	dBc/Hz	–

6.10 USB 2.0 PHY specification

Table 23 USB 2.0 PHY HS TX pre-emphasis

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_USB2_4	USB_2_PHY_HS_TX_AMP_0011	USB 2.0 PHY HS TX amplitude for register setting = 0011	720	800	880	mV	–
SID_USB2_6	USB_2_PHY_HS_TX_AMP_0101	USB 2.0 PHY HS TX amplitude for register setting = 0101	756	840	924	mV	–
SID_USB2_8	USB_2_PHY_HS_TX_AMP_0111	USB 2.0 PHY HS TX amplitude for register setting = 0111	792	880	968	mV	–
SID_USB2_12	USB_2_PHY_HS_TX_AMP_1011	USB 2.0 PHY HS TX amplitude for register setting = 1011	864	960	1056	mV	–
SID_USB2_19	USB_2_PHY_HS_TX_PREMP_010	USB 2.0 PHY HS TX pre-emphasis for register setting = 010	0.83	1.5	1.78	dB	–
SID_USB2_21	USB_2_PHY_HS_TX_PREMP_100	USB 2.0 PHY HS TX pre-emphasis for register setting = 100	2.28	2.9	3.52	dB	–
SID_USB2_23	USB_2_PHY_HS_TX_PREMP_110	USB 2.0 PHY HS TX pre-emphasis for register setting = 110	3.23	4	4.6	dB	–

Table 24 USB 2.0 PHY HS TX pre-emphasis duration

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_USB2_24	USB_2_PHY_HS_TX_PREMP_D_00	USB 2.0 PHY HS TX pre-emphasis duration for register setting = 00	–	0	–	UI	0 = Disabled
SID_USB2_25	USB_2_PHY_HS_TX_PREMP_D_01	USB 2.0 PHY HS TX pre-emphasis duration for register setting = 01	0.2	0.25	0.3	UI	–
SID_USB2_26	USB_2_PHY_HS_TX_PREMP_D_10	USB 2.0 PHY HS TX pre-emphasis duration for register setting = 10	0.45	0.5	0.55	UI	–
SID_USB2_27	USB_2_PHY_HS_TX_PREMP_D_11	USB 2.0 PHY HS TX pre-emphasis duration for register setting = 11	0.7	0.75	0.8	UI	–

Electrical specifications

Table 25 USB 2.0 PHY HS TX slew rate

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_USB2_28	USB_2_PHY_HS_TX_SLRA_00	USB 2.0 PHY HS TX slew rate for register setting = 00	100	300	460	ps	Guaranteed by characterization
SID_USB2_29	USB_2_PHY_HS_TX_SLRA_01	USB 2.0 PHY HS TX slew rate for register setting = 01	–	375	560	ps	Guaranteed by characterization
SID_USB2_30	USB_2_PHY_HS_TX_SLRA_10	USB 2.0 PHY HS TX slew rate for register setting = 10	–	450	680	ps	Guaranteed by characterization
SID_USB2_31	USB_2_PHY_HS_TX_SLRA_11	USB 2.0 PHY HS TX slew rate for register setting = 11	–	525	800	ps	Guaranteed by characterization

Table 26 USB 2.0 PHY Host Disconnect Thresholds

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID408	USB_2_PHY_DISC_TH_0000	USB 2.0 PHY Host Disconnect Threshold for register setting = 0000	525	575	625	mV	%Δ USB 2.0 spec = 0.0 mV
SID412	USB_2_PHY_DISC_TH_0100	USB 2.0 PHY Host Disconnect Threshold for register setting = 0100	605	660	715	mV	%Δ USB 2.0 spec = 15.2 mV
SID416	USB_2_PHY_DISC_TH_1000	USB 2.0 PHY Host Disconnect Threshold for register setting = 1000	685	745	805	mV	%Δ USB 2.0 spec = 30.4 mV
SID420	USB_2_PHY_DISC_TH_1100	USB 2.0 PHY Host Disconnect Threshold for register setting = 1100	765	830	895	mV	%Δ USB 2.0 spec = 45.6 mV

Table 27 USB 2.0 PHY squelch detection thresholds

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_USB2_49	USB_2_PHY_SQ_DET_1	USB 2.0 PHY squelch detection threshold for setting 1, Register = 1011	130	155	195	mV	–
SID_USB2_51	USB_2_PHY_SQ_DET_3	USB 2.0 PHY squelch detection threshold for setting 3, Register = 1001	115	140	172.5	mV	–
SID_USB2_53	USB_2_PHY_SQ_DET_5	USB 2.0 PHY squelch detection threshold for setting 5, Register = 0111	100	125	150	mV	–
SID_USB2_55	USB_2_PHY_SQ_DET_7	USB 2.0 PHY squelch detection threshold for setting 7, Register = 0101	85	110	135	mV	–

Electrical specifications

Table 28 USB 2.0 PHY HS CTLE equalization

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_USB2_56	USB_2_PHY_HS_CTLE_000	USB 2.0 PHY HS CTLE equalization for register setting = 000	-0.2	0	0.2	dB	Typical value @ 240 MHz. Guaranteed by characterization
SID_USB2_58	USB_2_PHY_HS_CTLE_010	USB 2.0 PHY HS CTLE equalization for register setting = 010	0.7	1.1	1.75	dB	-
SID_USB2_60	USB_2_PHY_HS_CTLE_100	USB 2.0 PHY HS CTLE equalization for register setting = 100	1.6	2.2	2.9	dB	-
SID_USB2_62	USB_2_PHY_HS_CTLE_110	USB 2.0 PHY HS CTLE equalization for register setting = 110	2.8	3.4	4.4	dB	-

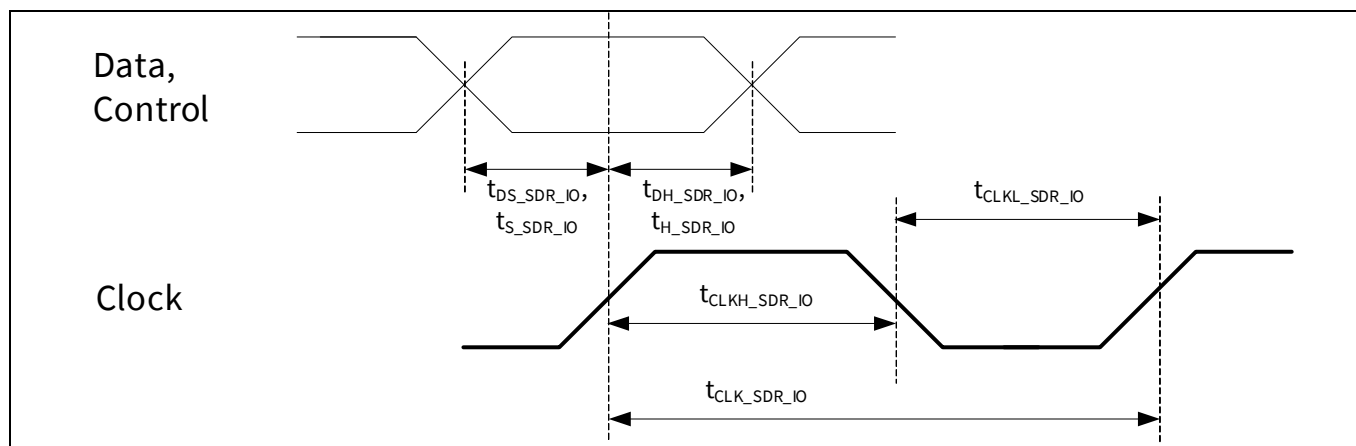
6.11 LVCMOS specification**Table 29 LVCMOS electrical parameter specifications**

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_LVCMOS1	VOH_LVCMOS	Output HIGH voltage	75	-	-	%VDDIO	-
SID_LVCMOS2	VOL_LVCMOS	Output LOW voltage	-	-	25	%VDDIO	-
SID_LVCMOS3	IIH_LVCMOS	Input HIGH current	-	-	1	μA	-
SID_LVCMOS4	IIL_LVCMOS	Input LOW current	-	-	1	μA	-
SID_LVCMOS5	VIH_LVCMOS	Input HIGH level	65	-	-	%VDDIO	-
SID_LVCMOS6	VIL_LVCMOS	Input LOW level	-	-	35	%VDDIO	-
SID_LVCMOS7	RHIGH_Z0	Pull-up resistance	40	-	60	Ω	-
SID_LVCMOS8	RLOW_Z0	Pull-down resistance	40	-	60	Ω	-
SID_LVCMOS9	IOH_DS0	Output HIGH current at VOH, lowest drive DS0	1.5	-	7	mA	-
SID_LVCMOS10	IOL_DS0	Output LOW current at VOL, lowest drive DS0	1.5	-	7	mA	-
SID_LVCMOS11	IOH_DS1	Output HIGH current at VOH, mid drive DS1	3.4	-	14	mA	-
SID_LVCMOS12	IOL_DS1	Output LOW current at VOL, mid drive DS1	3.4	-	14	mA	-
SID_LVCMOS13	IOH_DS2	Output HIGH current at VOH, mid drive DS2	5.2	-	20	mA	-
SID_LVCMOS14	IOL_DS2	Output LOW current at VOL, mid drive DS2	5.2	-	20	mA	-
SID_LVCMOS15	IOH_DS3	Output HIGH current at VOH, highest drive DS3	7	-	30	mA	-
SID_LVCMOS16	IOL_DS3	Output LOW current at VOL, highest drive DS3	7	-	30	mA	-

Electrical specifications

Table 30 LVCMOS SDR IO interface timing specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_LVCMOS36	fLVCMOS_SDR_IO	Clock frequency, when interface is used as I/O	–	–	100	MHz	–
SID_LVCMOS37	tCLK_SDR_IO	Clock period, when interface is used as I/O	10	–	–	ns	–
SID_LVCMOS38	tCLKH_SDR_IO	Clock high time, when interface is used as I/O	4	–	–	ns	–
SID_LVCMOS39	tCLKL_SDR_IO	Clock low time, when interface is used as I/O	4	–	–	ns	–
SID_LVCMOS40	tS_SDR_IO	CTL Input to clock setup Time, when interface is used as I/O	2	–	–	ns	–
SID_LVCMOS41	tH_SDR_IO	CTL Input to clock hold Time, when interface is used as I/O	0.5	–	–	ns	–
SID_LVCMOS42	tDS_SDR_IO	Data in to clock setup time, when interface is used as I/O	2	–	–	ns	–
SID_LVCMOS43	tDH_SDR_IO	Data in to clock hold time, when interface is used as I/O	0.5	–	–	ns	–
SID_LVCMOS44	tDO_SDR_IO	Clock to data out propagation delay when DQ bus is already in output direction	–	–	8	ns	–
SID_LVCMOS45	tCOE_SDR_IO	Clock to data out propagation delay when DQ lines change to output from tristate and valid data is available on the DQ bus	–	–	8	ns	–
SID_LVCMOS46	tCO_SDR_IO	Clock to CTL out propagation delay	–	–	8	ns	–
SID_LVCMOS47	tDOH_SDR_IO	Clock to data out hold	2	–	–	ns	–
SID_LVCMOS48	tCOH_SDR_IO	Clock to CTL out hold	2	–	–	ns	–
SID_LVCMOS49	tHZ_SDR_IO	Clock to High-Z	–	–	8	ns	–
SID_LVCMOS50	tLZ_SDR_IO	Clock to Low-Z	0	–	–	ns	–

**Figure 6** LVCMOS SDR IO clock and data timing

6.12 QSPI specifications

Table 31 QSPI specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID_QSPI1	Fsmifclk	SMIF QSPI output clock frequency	–	–	80	MHz	–
SID_QSPI3	Tsetup	Input data set-up time with respect to clock capturing edge	4.5	–	–	ns	–
SID_QSPI4	Tdatahold	Input data hold time with respect to clock capturing edge	0	–	–	ns	–
SID_QSPI5	Tdataoutvalid	Output data valid time with respect to clock falling edge	–	–	3.7	ns	–
SID_QSPI6	Tholdtime	Output data hold time with respect to clock rising edge	3	–	–	ns	–
SID_QSPI7	Tseloutvalid	Output select valid time with respect to clock rising edge	–	–	7.5	ns	–
SID_QSPI8	Tselouthold	Output select hold time with respect to clock rising edge	$0.5 \times T_{sclk}$	–	–	ns	$T_{sclk} = F_{smifclk}$ cycle time
SID_QSPI9	Tckpw	Clock pulse width	45	–	55	%	Related to Fsmifclk

6.13 Digital peripherals specifications

Table 32 TCPWM specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID_TCPWM_1	TCPWMFREQ	Operating frequency	–	–	100	MHz	F _c max = F _{cpu} . Maximum = 100 MHz
SID_TCPWM_2	TPWMENEXT	Input trigger pulse width for all trigger events	2/F _c	–	–	ns	Trigger Events can be Stop, Start, Reload, Count, Capture, or Kill depending on which mode of operation is selected
SID_TCPWM_3	TPWMEXT	Output trigger pulse widths	1.5/F _c	–	–	ns	Minimum possible width of Overflow, Underflow, and CC (Counter equals Compare value) trigger outputs
SID_TCPWM_3A	TCRES	Resolution of counter	1/F _c	–	–	ns	Minimum time between successive counts
SID_TCPWM_3B	PWMRES	PWM resolution	1/F _c	–	–	ns	Minimum pulse width of PWM Output
SID_TCPWM_3C	QRES	Quadrature inputs resolution	2/F _c	–	–	ns	Minimum pulse width between Quadrature phase inputs. Delays from pins should be similar.

Table 33 UART AC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID_UART1	FUART2	Bit rate	–	–	8	Mbps	–

Table 34 SPI AC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID_SPI1	FSPI	SPI operating frequency master	–	–	25	MHz	–
SID_SPI1A	FSPI_EXT	SPI operating frequency master (F _{scb} is SPI clock)	–	–	F _{scb} /4	MHz	F _{scb} max is 100 MHz
SID_SPI1B	FSPI_IC	SPI slave internally clocked	–	–	15	MHz	–

Electrical specifications

Table 35 SPI master mode AC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_SPIM1	TDMO	MOSI valid after Sclock driving edge	–	–	12	ns	–
SID_SPIM2	TDSI	MISO valid before Sclock capturing edge	20	–	–	ns	Full clock, late MISO sampling
SID_SPIM3	THMO	MOSI data hold time	0	–	–	ns	Referred to slave capturing edge
SID_SPIM3A	TDHI	SPI master: MISO hold time after SCLK capturing edge	0	–	–	ns	–
SID_SPIM3B	TSSELMCK1	SSEL valid to first SCK valid edge	18	–	–	ns	Referred to master clock edge
SID_SPIM3C	TSSELMCK2	SSEL valid to first SCK valid edge	18	–	–	ns	Referred to master clock edge

Table 36 SPI slave mode AC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_SPIS1	TDMI	MOSI valid before Sclock capturing edge	5	–	–	ns	–
SID_SPIS1A	SPI_FREQ	SPI frequency	–	–	25	MHz	–
SID_SPIS2	TDSO_EXT	MISO valid after Sclock driving edge in Ext. Clk. mode	–	–	20	ns	–
SID_SPIS2A	TDSO	MISO valid after Sclock driving edge in Internally Clk. mode	–	–	$TDSO_EXT + 3 \times Tscb$	ns	Tscb is serial comm block clock period
SID_SPIS2B	TDSO_FILT	MISO valid after Sclock driving edge in Internally Clk. mode with Median filter enabled.	–	–	$TDSO_EXT + 4 \times Tscb$	ns	Tscb is serial comm block clock period
SID_SPIS3	THSO	Previous MISO data hold time	5.5	–	–	ns	–
SID_SPIS3A	THIS	SPI MOSI hold from SCLK	–	–	14	ns	–
SID_SPIS3B	CSPI	SPI capacitive load	–	10	–	pF	–
SID_SPIS3C	TSSELSCK1	SSEL valid to first SCK valid edge	65	–	–	ns	–
SID_SPIS3D	TSSELSCK2	SSEL Hold after last SCK valid edge	65	–	–	ns	–

Electrical specifications

6.14 Audio subsystem specifications**Table 37 PDM specifications**

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID_PDM1	Fmax_clk_if_srss	Clock frequency for audio clock reference clk_if_srss	–	48	–	MHz	–
SID_PDM2	T_SETUP	Receiver setup	–	–	10	ns	–
SID_PDM3	PDM_HOLD	Data input hold time to PDM_CLK edge	10	–	–	ns	–
SID_PDM4	CPDM	Load	–	10	–	pF	–
SID_PDM5	PDM_OUT	Audio sample rate	8	–	48	ksps	–
SID_PDM6	PDM_WL	Word length	16	–	24	bits	–
SID_PDM7	PDM_SNR	Signal-to-noise ratio (A-weighted)	–	100	–	dB	PDM input, 20 Hz to 20 kHz BW
SID_PDM8	PDM_DR	Dynamic Range (A-weighted)	–	100	–	dB	20 Hz to 20 kHz BW, -60 dB FS
SID_PDM10	PDM_SB	Stop band	–	0.566	–	f	DC to 0.45 f. DC blocking filter OFF
SID_PDM11	PDM_SBA	Stop band attenuation	–	60	–	dB	–
SID_PDM12	PDM_GAIN	Adjustable gain	–12	–	10.5	dB	–
SID_PDM13	PDM_ST	Start-up time	–	48	–	WS cycles	PDM to PCM, 1.5 dB/step

6.15 JTAG specifications**Table 38 JTAG specifications**

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/conditions
SID_JTAG1	TCKLOW	TCK LOW minimum	34	–	–	ns	–
SID_JTAG2	TCKHIGH	TCK HIGH	10	–	–	ns	–
SID_JTAG3	TCKPERIOD	CLK_JTAG_PERIOD, 30 pF load	–	44	–	MHz	–
SID_JTAG4	TCK_TDO	TDO clock-to-out (maximum) from falling TCK	–	–	22	ns	–
SID_JTAG5	TSU_TCK	TDI, TMS setup time before rising TCK	12	–	–	ns	–
SID_JTAG6	TCK_THD	TDI, TMS hold time after rising TCK	10	–	–	ns	–
SID_JTAG7	TCK_TDOV	TCK to TDO data valid (High-Z to active)	22	–	–	ns	–
SID_JTAG8	TCK_TDOZ	TCK to TDO data valid (active to High-Z)	22	–	–	ns	–
SID_JTAG9	JTAG_TDO_HOLD	JTAG TDO hold time	–	–	5	ns	–
SID_JTAG10	JTAG_INPUT_TRANSITION_TIME	JTAG input transition time	–	–	5	ns	–

Electrical specifications

6.16 SSRS specifications

Table 39 IMO AC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_IMO1	FIMOTOL1	Frequency variation centered on 8 MHz	-2	-	2	%	-
SID_IMO2	TJITR	Cycle-to-cycle and period jitter	-	-	250	ps	-

Table 40 ILO AC specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_ILO1	TSTARTILO1	ILO start-up time	-	-	7	μs	Start-up time to 95% of final frequency
SID_ILO2	TLIODUTY	ILO duty cycle	45	50	55	%	-
SID_ILO3	FILOTRIM1	32 kHz trimmed frequency	-10	-	10	%	28.8 kHz to 35.2 kHz (±10% variation)

Table 41 Frequency locked loop (FLL) specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_FLL1	FLL_RANGE	Input frequency range	0.04	-	96	MHz	Upper limit is for external input
SID_FLL2	FLL_OUT_DIV2	Output frequency range	24	-	96	MHz	Output range of FLL divided-by-2 output
SID_FLL4	FLL_DUTY_DIV2	Divided-by-2 output; HIGH or LOW	47	-	53	%	-
SID_FLL5	FLL_WAKEUP	Time from stable input clock to 1% of final value on Deep Sleep wakeup	-	-	11	μs	With IMO input, for <10°C change in temperature while in Deep Sleep and Fout ≥ 50 MHz
SID_FLL6	FLL_JITTER	Period jitter (1 sigma) at 96 MHz	-	-	45	ps	-

Table 42 External clock specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_ILO1	TSTARTILO1	ILO Start-up time	-	-	7	μs	-
SID_ILO2	TLIODUTY	ILO Duty cycle	45	50	55	%	-
SID_ILO3	FILOTRIM1	32 kHz trimmed frequency	-10	-	10	%	-

6.17 Flash specification

Table 43 Flash specifications

Spec ID	Parameter	Description	Min	Typ	Max	Unit	Details/ conditions
SID_FLASH1	TROWWRITE	Row write item (erase and program)	–	–	16	ms	–
SID_FLASH2	TROWERASE	Row erase time	–	–	11	ms	–
SID_FLASH3	TROWPROGRAM	Row program time after erase	–	–	5	ms	–
SID_FLASH4	TBULKERASE	Bulk erase time (2048 KB)	–	–	11	ms	–
SID_FLASH5	TSECTORERASE	Sector erase time (256 KB)	–	–	11	ms	–
SID_FLASH6	TSSERIAE	Sub-sector erase time	–	–	11	ms	–
SID_FLASH7	TSSWRITE	Sub-sector erase time; 1 erase plus 8 program times	–	–	51	ms	–
SID_FLASH7	TSWRITE	Total device write time	–	–	2.6	s	–
SID_FLASH8	TDEVPROG	Sub-sector write time; 1 erase plus 512 program times	–	–	30	s	–
SID_FLASH9	FEND	Flash endurance	100K	–	–	cycles	–
SID_FLASH10	FRET1	Flash retention. TA ≤ 25°C, 100K P/E cycles	10	–	–	yrs	–
SID_FLASH11	FRET2	Flash retention. TA ≤ 85°C, 10K P/E cycles	10	–	–	yrs	–
SID_FLASH12	FRET3	Flash retention. TA ≤ 55°C, 20K P/E cycles	20	–	–	yrs	–
SID_FLASH13	TWS100	Number of wait states at 100 MHz	300%	–	–	–	–

7 Timing and electrical diagrams

7.1 LVCMOS SDR IO synchronous mode GPIF III timing

This section describes the generic slave FIFO interface diagrams.

7.1.1 LVCMOS SDR IO GPIF III interface example

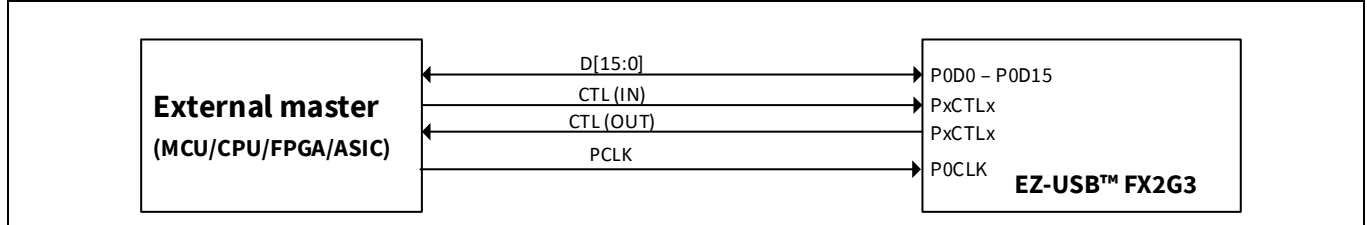


Figure 7 LVCMOS SDR IO GPIF III interface example

7.1.2 GPIF III timing in LVCMOS SDR IO synchronous mode

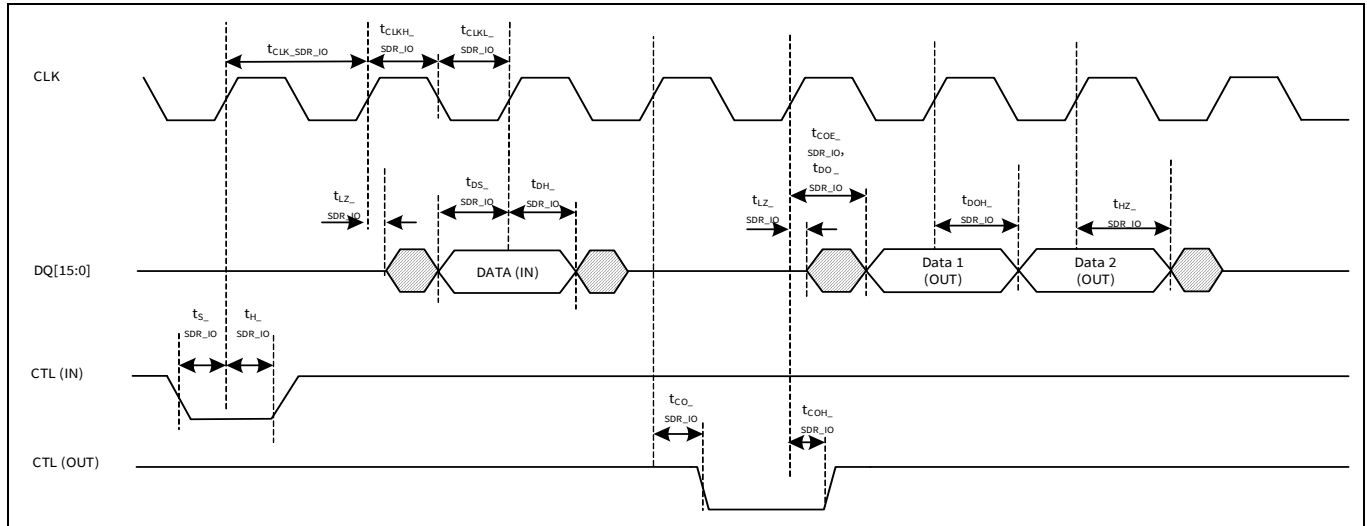


Figure 8 GPIF III timing in LVCMOS SDR IO synchronous mode (see Table 30)

7.1.3 Slave FIFO write interface example

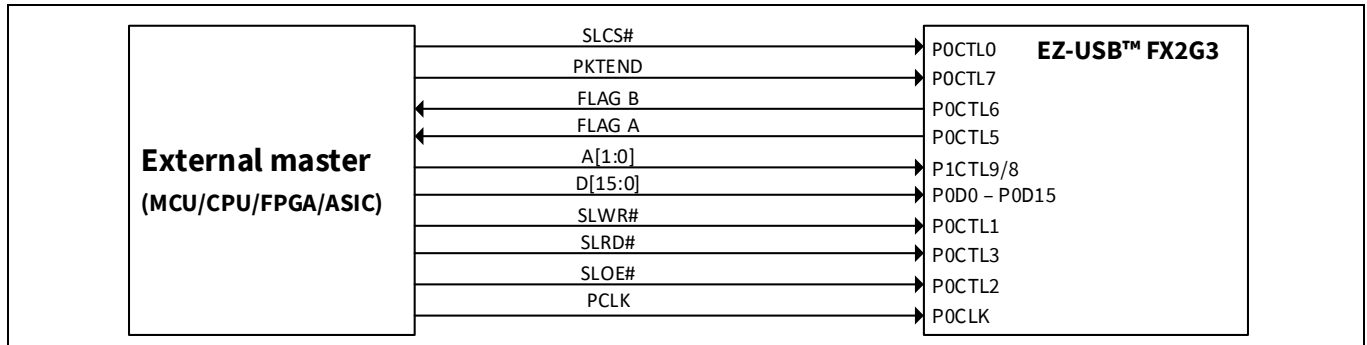
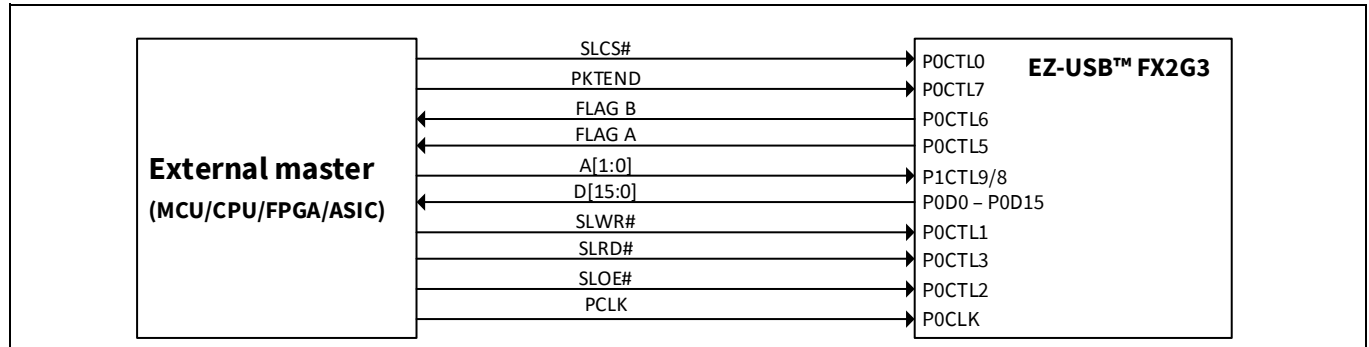


Figure 9 Slave FIFO write interface example

7.1.4 Slave FIFO read interface example



7.1.5 Synchronous SDR slave FIFO read mode

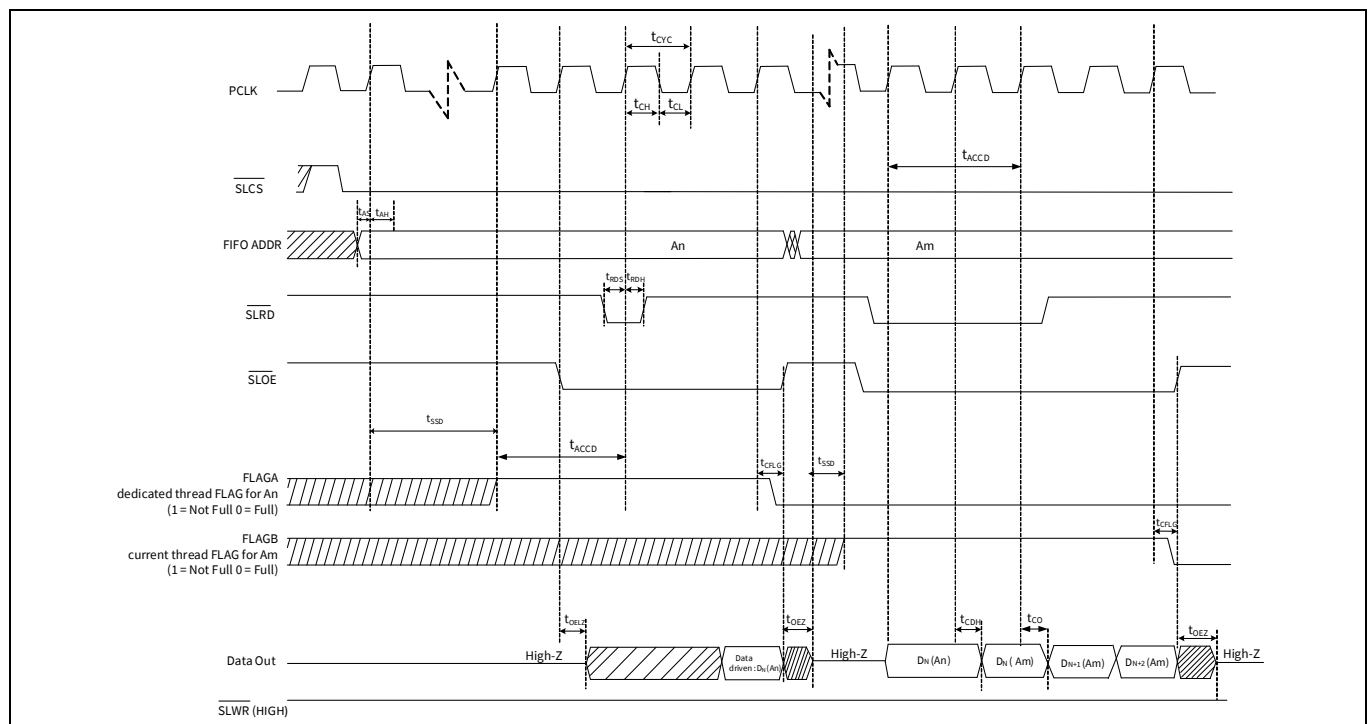


Figure 10 Synchronous SDR slave FIFO read cycle timing

7.1.6 Synchronous SDR slave FIFO write mode

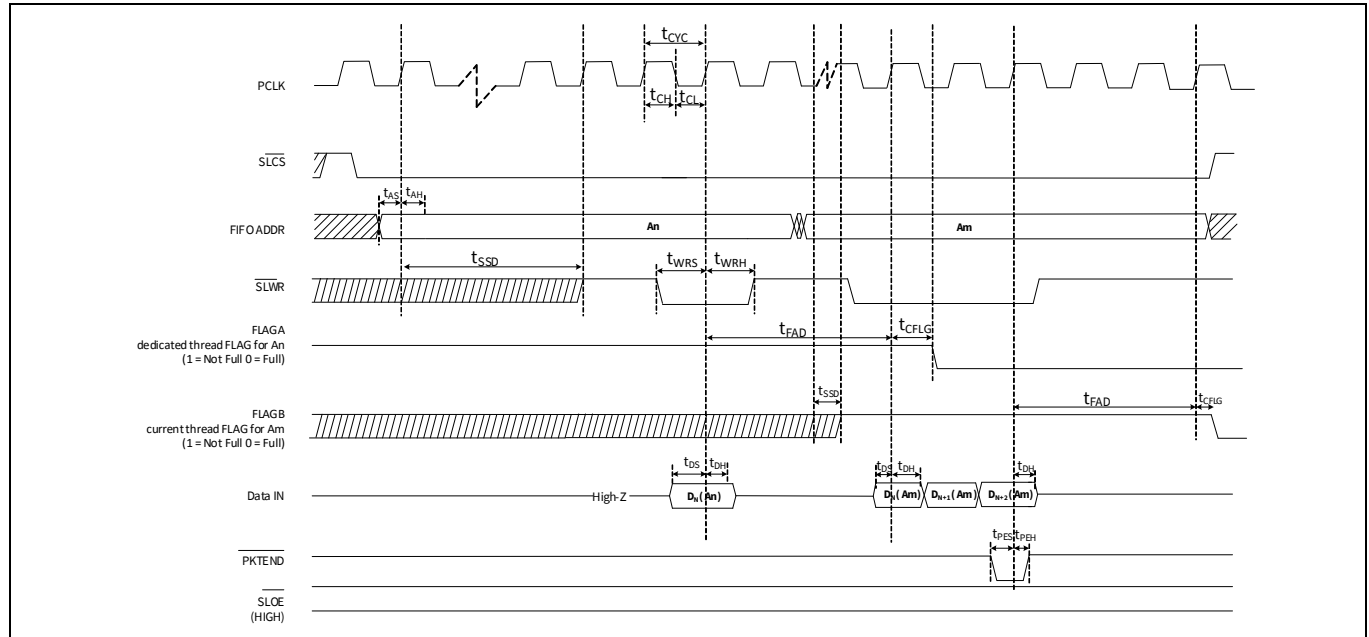


Figure 11 Synchronous SDR slave FIFO write cycle timing

7.1.7 Synchronous SDR slave FIFO ZLP write mode

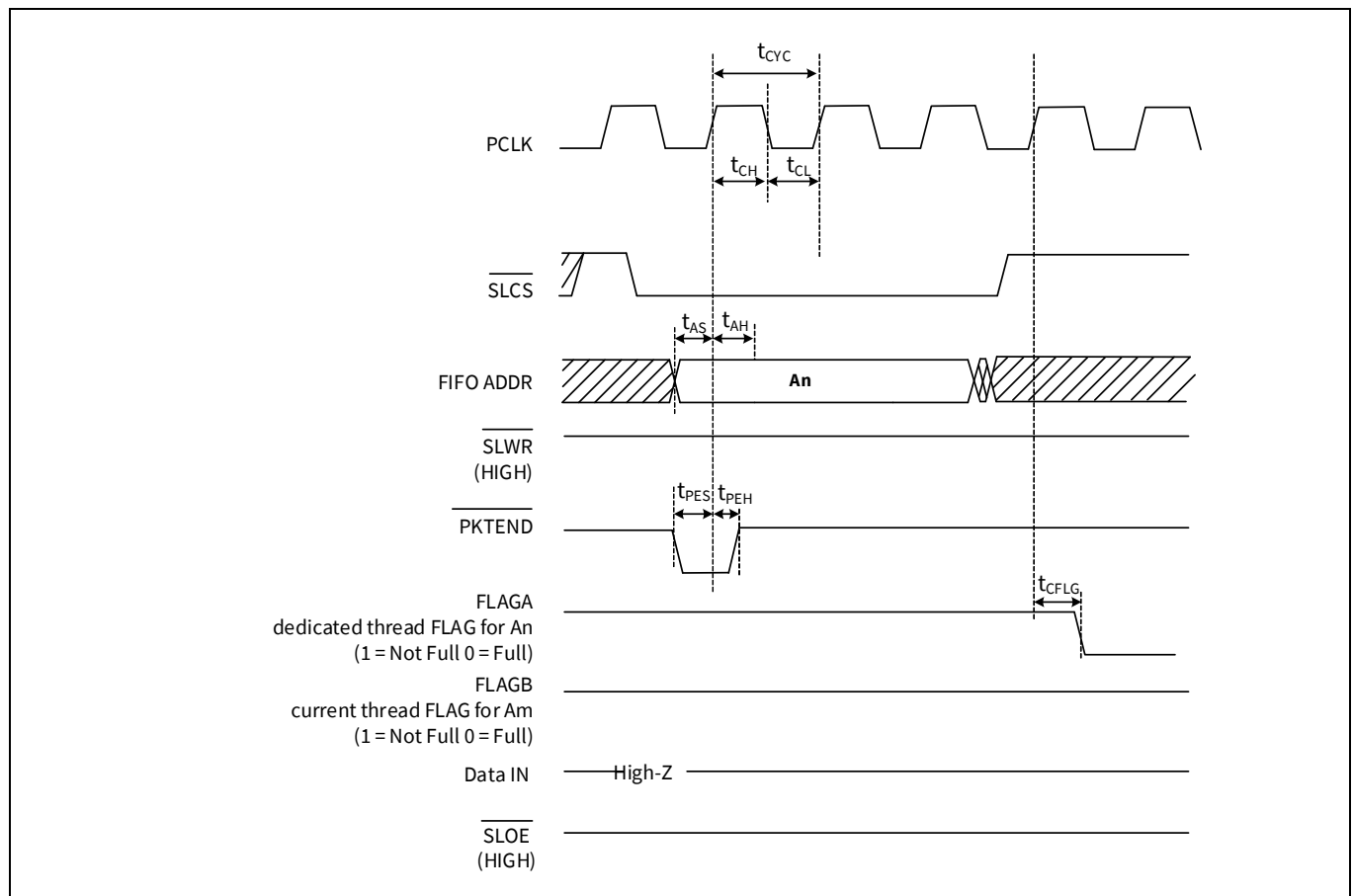


Figure 12 Synchronous SDR slave FIFO ZLP write cycle timing

Table 44 Synchronous SDR slave FIFO parameters^[3]

Parameter	Description	Reference mapping	Min	Max	Unit
FREQ	Interface clock frequency	fLVCMOS_SDR_IO	–	100	MHz
tCYC	Clock period	tCLK_SDR_IO	10	–	ns
tCH	Clock high time+	tCLKH_SDR_IO	4	–	ns
tCL	Clock low time	tCLKL_SDR_IO	4	–	ns
tRDS	SLRD# to CLK setup time	tS_SDR_IO	2	–	ns
tRDH	SLRD# to CLK hold time	tH_SDR_IO	0.5	–	ns
tWRS	SLWR# to CLK setup time	tS_SDR_IO	2	–	ns
tWRH	SLWR# to CLK hold time	tH_SDR_IO	0.5	–	ns
tCO	Clock to valid data	tDO_SDR_IO	–	8	ns
tDS	Data input setup time	tDS_SDR_IO	2	–	ns
tDH	CLK to data input hold	tDH_SDR_IO	0.5	–	ns
tAS	Address to CLK setup time	tDS_SDR_IO	2	–	ns
tAH	CLK to address hold time	tDH_SDR_IO	0.5	–	ns
tOELZ	SLOE# to data Low-Z	tLZ_SDR_IO	0	–	ns
tCFLG	CLK to flag output propagation delay	tCO_SDR_IO	–	8	ns
tOEZ	SLOE# deassert to Data high-Z	tHZ_SDR_IO	–	8	ns
tPES	PKTEND# to CLK setup	tS_SDR_IO	2	–	ns
tPEH	CLK to PKTEND# hold	tH_SDR_IO	0.5	–	ns
tCDH	CLK to data output hold	tDOH_SDR_IO	2	–	ns
tSSD	Socket switching delay	–	2	68	Clock cycles
tACCD	Latency from SLRD# to Data	–	2	2	Clock cycles
tFAD	Latency from SLWR# to FLAG	–	3	3	Clock cycles

Note Three-cycle latency from ADDR to DATA/FLAGS.

Note

3. All parameters guaranteed by design and validated through characterization.

Table 45 **Slave FIFO pin mapping**

Internal Signals	Narrow link 0		
	8-bit wide bus		16-bit bus
	2-bit FIFO	5-bit FIFO	2-bit FIFO
CLK	CLK	CLK	CLK
DQ[0]	D0	D0	D0
DQ[1]	D1	D1	D1
DQ[2]	D2	D2	D2
DQ[3]	D3	D3	D3
DQ[4]	D4	D4	D4
DQ[5]	D5	D5	D5
DQ[6]	D6	D6	D6
DQ[7]	D7	D7	D7
DQ[8]	A0	A0	D8
DQ[9]	A1	A1	D9
DQ[10]	SPARE	A2	D10
DQ[11]	SPARE	A3	D11
DQ[12]	SPARE	A4	D12
DQ[13]	SPARE	SPARE	D13
DQ[14]	SPARE	SPARE	D14
DQ[15]	SPARE	SPARE	D15
CTL[9]	SPARE	SPARE	A0
CTL[8]	SPARE	EPSWITCH	A1
CTL[7]	L3_ENTRY	L3_ENTRY	L3_ENTRY
CTL[6]	FLAG B	FLAG B	FLAG B
CTL[5]	FLAG A	FLAG A	FLAG A
CTL[4]	PKTEND	PKTEND	PKTEND
CTL[3]	RD	RD	RD
CTL[2]	OE	OE	OE
CTL[1]	WE	WE	WE
CTL[0]	CE	CE	CE

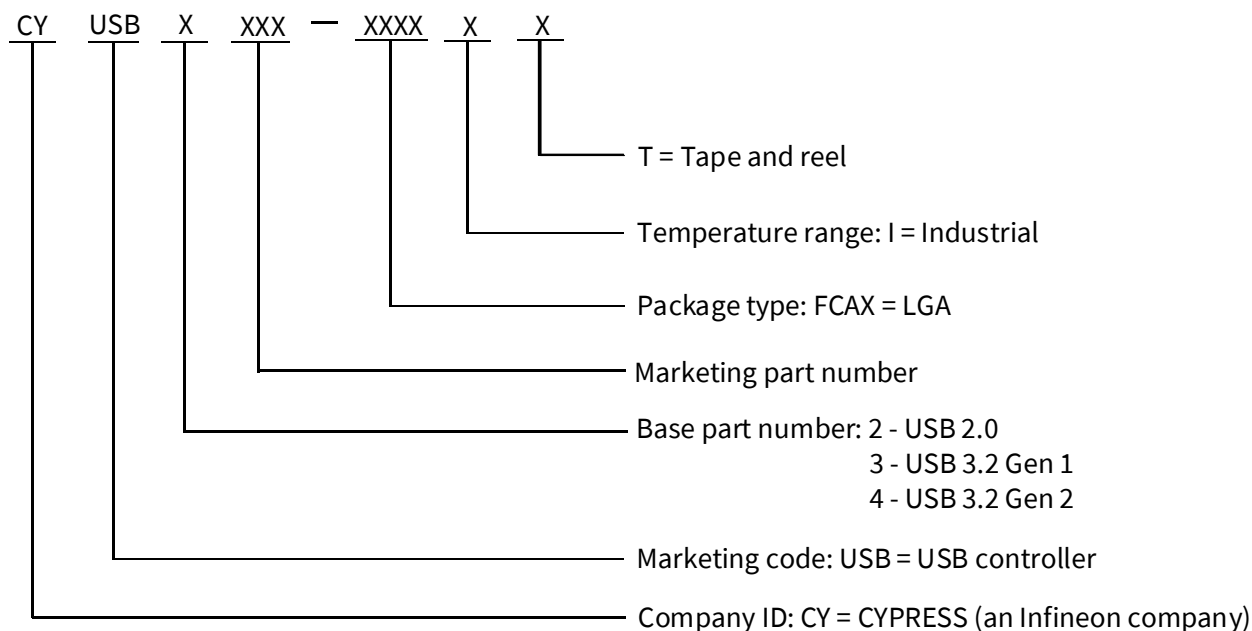
8 Ordering information

Table 45 lists the EZ-USB™ FX2G3 key package features and ordering codes. The table contains only the parts that are currently available. If you do not see what you are looking for, contact your local sales representative.

Table 46 EZ-USB™ FX2G3 key features and ordering information

Product	Package	Temp (°C)	GPIF/FIFO	SRAM (KB)	MCU	I2C	FLASH (KB)	Crypto	QSPI	SPI UART CAN
CYUSB2315-BF104AXI	104 LGA	–45 to 85	16-bit	256	M0+	Yes	256			
CYUSB2316-BF104AXI	104 LGA	–45 to 85	16-bit	256	M0+	Yes	256			Yes
CYUSB2317-BF104AXI	104 LGA	–45 to 85	16-bit	512	M0+	Yes	512		Yes	Yes
CYUSB2318-BF104AXI	104 LGA	–45 to 85	16-bit	1024	M4/M0+	Yes	512	Yes	Yes	Yes

8.1 Ordering code definitions



9 Package diagram

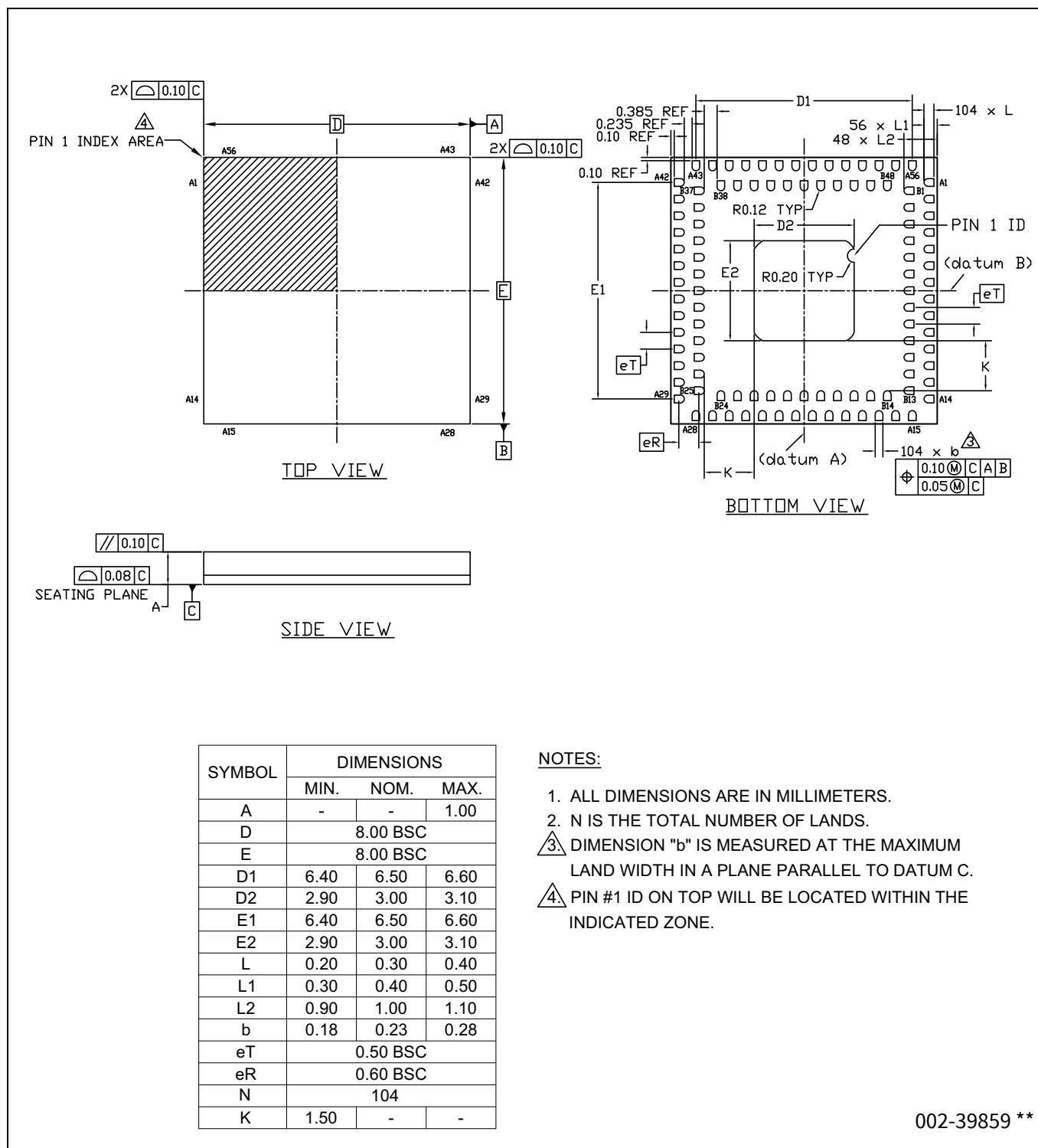


Figure 13 104 lead LGA 8.00 × 8.00 × 1.00 mm VDB104 package outline

10 Acronyms

Table 47 Acronyms used in this document

Acronym	Description
ADC	analog-to-digital converter
AES	advanced encryption standard
AHB	AMBA high-performance bus
AXI	advanced eXtensible interface
BB	bit banding
BOD	brown-out detection
BSDL	boundary scan description language
CAN	controller area network
CRC	cyclic redundancy check
CDC	communication device class
CRYPTO	cryptography accelerator
CTI	cross trigger interface
CTR	counter
DFT	design for testability
DMA	direct memory access
DW	data width
ECC	elliptic curve cryptography
ETM	embedded trace macrocell
FBGA	fine ball grid array
FLL	frequency-locked loop
FPU	floating point unit
FS	full-speed
GPIF	general programmable interface
GPIO	general-purpose I/O
HID	human-interface device
HS	high-speed
I2C	inter integrated circuit
IBIS	input output buffer information system
ILO	internal low-speed oscillator
IMO	internal main oscillator
IOSS	input output subsystem
IPC	inter-processor communication
ITM	instrumentation trace macrocell
LDO	low dropout regulator
LP	low power
LVD	low voltage detection
LVC MOS	low voltage complementary metal oxide semiconductor
MCU	microcontroller unit
MPU	memory protection unit
MMIO	memory-mapped IO
MTB	micro trace buffer

Acronyms

Table 47 **Acronyms used in this document** *(continued)*

Acronym	Description
NVIC	nested vector interrupt control
OVP	overvoltage protection
PCLK	peripheral clock
PCM	pulse code modulation
PDM	pulse density modulation
PHY	physical layer
PLL	phase-locked loop
POR	power-on reset
PPU	power policy unit
PWM	pulse-width modulation
PWRSYS	power system
QD	quadrature decoder
QSPI	quad SPI
REF	reference voltage
RSA	Rivest–Shamir–Adleman cryptography algorithm
RWW	read-while-write
SCB	serial communication block
SHA	secure hash algorithm
SMPU	system memory protection unit
SPI	serial peripheral interface
SRAM	static random access memory
SWJ	SWD or JTAG debug port
TOF	time-of-flight
TRNG	true random number generation
UAC	USB audio class
UART	universal asynchronous receiver transmitter
USB	Universal Serial Bus
UVC	USB video class
WDT	watchdog timer
XRES	external reset

11 Document conventions

11.1 Unit of measure

Table 48 Unit of measure

Symbol	Unit of measure
°C	degrees celsius
dB	decibel
fF	femto farad
Gbps	gigabits per second
Hz	hertz
KB	1024 bytes
kbps	kilobits per second
kHz	kilohertz
kΩ	kilo-ohm
ksps	kilosamples per second
LSB	least significant bit
Mbps	megabits per second
MHz	megahertz
MΩ	mega-ohm
Msps	megasamples per second
μA	microampere
μF	microfarad
μH	microhenry
μs	microsecond
μV	microvolt
μW	microwatt
mA	milliampere
ms	millisecond
mV	millivolt
nA	nanoampere
ns	nanosecond
nV	nanovolt
Ω	ohm
pF	picofarad
ppm	parts per million
ps	picosecond
s	second
sps	samples per second
sqrtHz	square root of hertz
V	volt



Revision history

Document revision	Date	Description of changes
**	2024-06-03	Initial release.
*A	2024-09-19	Added Table 26 . Updated Table 8 , Table 13 , and Table 14 . Updated Table 46 Removed P1D0 and P1D15 from figure in Slave FIFO read interface example . Updated number of SCBs in Features and Serial Communication Block (SCB) . Updated package information in Features . Updated Block diagram . Updated Table 1 . Updated GPIF III interface . Added Table 45 .

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